

2026 Graduation Project Final Report

Effect of Biomedical Knowledge Graph
Composition
on R-GCN-Based Drug–Drug Interaction
Prediction



Team CHEERS

Project No.: 05 Division: A – Software / Artificial Intelligence

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Abstract

Drug–drug interaction (DDI) prediction is a link-prediction problem in which model behavior depends not only on the learning architecture but also on the structure and semantics of the graph supplied to it. This project investigates the controlled question: *how does biomedical knowledge-graph composition affect R-GCN-based DDI link prediction?* Using PrimeKG as the final model-training source, the symmetric PrimeKG `drug_drug` relation is canonicalized into 1,336,314 unique undirected DDI pairs and divided into a fixed transductive split of 1,069,080 training pairs, 133,620 validation pairs, and 133,614 test pairs. Four graph variants are compared while holding the target split, candidate drug set, R-GCN architecture, decoder, optimization procedure, and ranking protocol fixed: **G0** contains DDI edges only; **G1** adds Drug–Gene/Protein relations; **G2** adds Drug–Disease relations; and **G3** combines both biomedical relation families.

The final model is a two-layer R-GCN with 128-dimensional representations and a symmetric DistMult-style DDI decoder. Across five training seeds, **G3** obtains the highest descriptive mean MRR, 0.534209 ± 0.006288 , compared with **G0** at 0.527284 ± 0.006373 . G3 exceeds G0 descriptively for all four reported ranking metrics across the five matched seeds, but the paired 95% intervals include zero; the improvement is therefore not statistically established under the fixed-split five-seed protocol. Five-seed single-relation ablation similarly finds no individual relation with an established positive effect. Complementary analyses examine thresholded classification, DDI-edge cold start, exploratory DDInter transfer, and qualitative graph context.

The verified G3/seed-44 model is integrated into a FastAPI and React/Vite research application using a lightweight NumPy inference export. Graph explorers expose G3 neighborhood context and grounded entity metadata, while openFDA label retrieval, PubMed literature retrieval, browser-side OCR, NCBI Gene metadata, and disease-information features provide independent application-level evidence or context and do not alter the R-GCN score. CHEERS is a research prototype rather than a clinical decision-support system; its raw model outputs are ranking scores, not probabilities, clinical risk estimates, DDI severity, or treatment recommendations.

Keywords: drug–drug interaction, knowledge graph, PrimeKG, R-GCN, link prediction, relation ablation, cold start, biomedical graph learning

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Contents

Abstract	i
Acknowledgements	ii
List of Abbreviations	viii
1 Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement	1
1.3 Research Questions	2
1.4 Research Objectives	2
1.5 Contributions	3
1.6 Scope and Responsible Interpretation	3
1.7 Project Evolution	3
2 Background and Related Work	4
2.1 Drug-Drug Interaction Prediction	4
2.2 Biomedical Knowledge Graphs	4
2.3 PrimeKG	5
2.4 Existing DDI Resources and Related Prediction Methods	5
2.5 Knowledge Graph Embedding Models	6
2.6 Graph Neural Networks and Message Passing	7
2.7 Relational Graph Convolutional Networks	7
2.8 DistMult-Style Bilinear Decoding	7
2.9 Link Prediction	8
2.10 Transductive and Inductive Evaluation	8
2.11 Relation-Level Ablation	8
2.12 Cold-Start Evaluation	8
2.13 Evaluation Metrics	8
2.14 Paired Statistical Analysis Across Seeds	9
3 Dataset and Knowledge Graph Construction	9
3.1 Final Data-Source Scope	9
3.2 PrimeKG DDI Extraction	10
3.3 Canonical DDI Split	10
3.4 Leakage Safeguards	11
3.5 Transductive Setting	11
3.6 Biomedical Support Relations	11
3.7 Graph Variants	11
3.8 Global Relation Mapping	13
3.9 Purpose of Reverse Support Relations	14
3.10 Controlled Experimental Variable	14
4 R-GCN Methodology	15
4.1 Overall Model Pipeline	15
4.2 Learned Node Embeddings	16
4.3 Relational Graph Convolution	16

4.4	Activation and Regularization	16
4.5	Symmetric DDI Decoder	16
4.6	Training Objective	16
4.7	Negative Sampling	17
4.8	Optimization and Hyperparameters	17
4.9	Model Selection	18
4.10	Training Environment	18
4.11	Filtered Ranking Evaluation	18
4.12	Complementary Classification Evaluation	19
5	Main Experimental Results	19
5.1	Five-Seed Graph-Composition Comparison	19
5.2	Matched G0 and G3 Seed Results	20
5.3	Complementary Classification Results	20
5.4	Paired Statistical Validation	20
5.5	Interpretation of “Interval Includes Zero”	21
5.6	Why G2 Matters	21
6	Relation-Level Ablation	21
6.1	Motivation	21
6.2	Ablation Construction	21
6.3	Five-Seed Ablation Results	22
6.4	Uncertainty	22
6.5	Relation Size Does Not Determine Contribution	23
6.6	Ranking Versus Classification	23
6.7	Ablation Conclusion	23
7	Generalization and External Evaluation	23
7.1	Purpose	23
7.2	DDI-Edge Cold-Start Definition	23
7.3	Cold-Drug Cohort	24
7.4	Cold-to-Warm Results	24
7.5	Cold-to-Cold Results	24
7.6	Cold-Start Interpretation	25
7.7	DDInter External Evaluation	25
7.8	External Ranking Results	26
7.9	Limitations of External Evaluation	26
7.10	Qualitative Case Studies	26
8	CHEERS System Architecture and Implementation	27
8.1	Purpose of the Research Application	27
8.2	Architecture Overview	27
8.3	Selected Demonstration Model	28
8.4	Lightweight NumPy Runtime	29
8.5	Runtime Verification	29
8.6	FastAPI Backend	29
8.7	React/Vite Frontend	30
8.8	Public Home and Search Interface	30

8.9	DDI Predictor	31
8.10	Pair Graph Explorer	31
8.11	Single-Drug Subgraph Explorer	32
8.12	Entity Details and Grounded Metadata	33
8.13	Independent FDA and PubMed Evidence	34
8.14	Medicine Label OCR	34
8.15	Disease and Nutrition Information	35
8.16	Research Methodology and Relation-Analysis Views	35
8.17	Responsible Interface Design	36
9	Verification and Reproducibility	37
9.1	Reproducibility Strategy	37
9.2	Versioned Result Artifacts	37
9.3	Checkpoints and Runtime Artifacts	37
9.4	Verification Scripts	37
9.5	Manifests and Checksums	38
9.6	Git-Based Development History	38
9.7	Portable Repository Versus University Workspace	38
10	Discussion	39
10.1	RQ1: Does Graph Composition Affect DDI Ranking?	39
10.2	RQ2: Which Relation Is Most Useful?	39
10.3	RQ3: Does Heterogeneous Context Help Cold Start?	39
10.4	Combined Context Versus Individual Relations	39
10.5	Ranking and Classification Measure Different Behavior	40
10.6	Statistical Uncertainty and Experimental Scale	40
10.7	External Generalization	40
10.8	Interpretability and Evidence Separation	40
10.9	Value of the Web Application	41
11	Project Limitations	41
12	Conclusion	42
13	Team Contributions	43

List of Tables

1	Comparison with representative DDI resources and graph-based prediction methods.	6
2	Role of principal data sources in the final CHEERS project	10
3	Canonical DDI split used by all final graph variants	10
4	Forward biomedical support-edge counts	11
5	Final graph variants	13
6	Global relation-ID mapping	14
7	Final R-GCN training configuration	17
8	Five-seed filtered-ranking results	20
9	Matched G0 and G3 MRR across seeds	20
10	Five-seed complementary classification performance	20
11	Paired G3–G0 statistical validation	20
12	Five-seed relation-ablation ranking results	22
13	Cold-to-warm filtered-ranking results	24
14	Cold-to-cold filtered-ranking results	24
15	Exploratory DDInter ranking results, seed 44	26
16	Principal CHEERS API endpoints	29

List of Figures

1	Controlled PrimeKG-derived graph-composition variants used in the final R-GCN experiment. G0 contains only drug–drug interaction edges; G1 adds Drug–Gene/Protein molecular context; G2 adds Drug–Disease context; and G3 combines both biomedical context families. The DDI split, node population, model architecture, decoder, and evaluation protocol are kept fixed across all variants.	12
2	Final CHEERS R-GCN link-prediction pipeline. Each node begins with a learned 128-dimensional embedding, followed by two relational graph-convolution layers with ReLU and dropout between them. Drug-pair representations are scored using a symmetric DistMult-style DDI decoder. The output is a ranking score, not a calibrated probability or clinical-risk value.	15
3	Five-seed MRR comparison across G0–G3. Error bars show sample standard deviation across seeds.	19
4	Five-seed paired relation-ablation MRR effects relative to G0. Every pointwise 95% paired interval crosses zero.	22
5	DDI-edge cold-start MRR comparison. The two evaluation conditions produce different conclusions.	25
6	Final CHEERS system architecture. The offline research pipeline constructs PrimeKG-derived graph variants, trains the R-GCN models, and exports the verified G3 seed-44 checkpoint for lightweight NumPy inference. The deployed application separates prediction, graph-context services, grounded metadata, independent FDA/PubMed evidence retrieval, and browser-side OCR. Evidence and context services support interpretation and navigation, but they do not alter the R-GCN ranking score.	28
7	CHEERS public home and search interface.	30
8	CHEERS DDI Predictor interface showing a selected query drug and ranked candidate-link workflow.	31
9	Pair-specific G3 context visualization for Warfarin and Tiagabine, with relation-aware node details.	32
10	Single-drug one-hop G3 Subgraph Explorer with relation and entity-type filters.	33
11	Independent evidence view for a drug pair. Official label excerpts and PubMed records are retrieved separately from R-GCN inference and do not alter or validate the model score.	34
12	Browser-side medicine-label OCR workflow after recognizing printed label text and conservatively matching Warfarin as a supported project drug. OCR is an input-selection aid, not clinical medicine identification.	35
13	CHEERS methodology page summarizing the fixed split, graph variants, R-GCN architecture, and evaluation design.	36
14	Five-seed relation-level ablation results shown in the research interface. All displayed 95% paired intervals include zero.	36

List of Abbreviations

DDI	Drug–Drug Interaction
KG	Knowledge Graph
KGE	Knowledge Graph Embedding
GNN	Graph Neural Network
R-GCN	Relational Graph Convolutional Network
MRR	Mean Reciprocal Rank
BCE	Binary Cross-Entropy
API	Application Programming Interface
OCR	Optical Character Recognition
FDA	U.S. Food and Drug Administration
NCBI	National Center for Biotechnology Information
UI	User Interface

1. Introduction

1.1 Background and Motivation

Drug–drug interactions (DDIs) are inherently relational phenomena: the interpretation of a drug pair can depend on the biological and therapeutic context surrounding each drug. Conventional DDI resources record known or curated interaction pairs, but a graph-learning system must reason over an incomplete set of observed relationships and rank plausible links that are not already represented as known positives. In such a setting, the graph supplied to the model is itself an experimental design choice. A DDI-only graph emphasizes observed drug connectivity, whereas a heterogeneous biomedical graph may additionally connect drugs to genes, proteins, diseases, indications, contraindications, transport processes, and related biomedical entities.

Knowledge graphs provide a natural representation for this problem because they encode entities as nodes and typed relationships as edges. PrimeKG was developed as a multimodal precision-medicine knowledge graph integrating numerous biomedical resources and relation types [1]. Its heterogeneous structure makes it possible to ask a question that is not answered by DDI-only modeling: whether non-DDI biomedical relations improve the ability of a graph neural network to rank held-out DDI links.

Relational Graph Convolutional Networks (R-GCNs) extend graph convolution to multi-relational data by assigning relation-dependent transformations during message passing [2]. They are therefore suitable for controlled comparison of graphs that share the same target relation but differ in their biomedical support relations. The central motivation of this project is not to assume that a larger or richer graph is automatically better, but to measure whether the composition of that graph changes predictive behavior under a fixed model and evaluation protocol.

1.2 Problem Statement

The final CHEERS project studies PrimeKG DDI link prediction under four controlled graph compositions. The target relation is PrimeKG’s `drug_drug` relation, displayed in the source data as a synergistic interaction relation. The model is given a graph built from training DDI edges and, depending on the experimental variant, selected Drug–Gene/Protein and Drug–Disease support relations. It then learns node representations and assigns a real-valued score to candidate drug pairs.

A high score is interpreted only as a high model ranking for an unobserved target link under the trained representation. It is not a probability of interaction, a calibrated confidence value, a clinical-risk score, an interaction-severity estimate, or evidence that a drug pair is safe or dangerous. This distinction is especially important because a knowledge graph is incomplete: absence of a recorded link does not establish the absence of a real-world interaction.

The central experimental problem is therefore:

How does biomedical knowledge-graph composition affect R-GCN-based drug–drug link prediction when the DDI split, node population, model architecture, decoder, training procedure, and evaluation protocol are held fixed?

1.3 Research Questions

The final project is organized around three research questions.

1.3.1 RQ1: Effect of Graph Composition

How does adding heterogeneous biomedical context to a DDI graph affect R-GCN filtered-ranking performance?

This question is addressed by the four graph variants **G0–G3**. The design isolates graph composition as the primary independent variable.

1.3.2 RQ2: Contribution of Individual Relations

Which individual biomedical relation types provide useful predictive information beyond a DDI-only graph?

This question is investigated using seven single-relation ablations that add exactly one biomedical relation family, with reverse edges for message passing, to the **G0** backbone.

1.3.3 RQ3: Generalization Under Reduced DDI Supervision

How does heterogeneous biomedical context affect prediction when selected drugs lose their DDI training/message-passing edges?

The implemented protocol is a *DDI-edge cold-start* experiment. It is not a fully inductive unseen-node experiment because the model retains learned node-ID embeddings and, in **G3**, selected cold drugs can retain non-DDI biomedical context.

1.4 Research Objectives

The final objectives are:

- 1) construct a leakage-controlled canonical DDI split from PrimeKG;
- 2) define four graph variants that differ only in biomedical relation composition;
- 3) train an identical R-GCN architecture across variants and five random seeds;
- 4) evaluate link prediction using full filtered ranking over the candidate drug set;
- 5) quantify seed-level uncertainty using paired analysis rather than relying only on mean metrics;
- 6) perform single-relation ablation to identify whether any individual support relation has a stable contribution;
- 7) evaluate DDI-edge cold-start behavior and conduct exploratory external evaluation on DDInter;
- 8) export a verified lightweight runtime for a research web application; and
- 9) separate model prediction from independent metadata, literature, label evidence, and educational content in the user interface.

1.5 Contributions

The project makes five principal contributions.

First, it defines a controlled PrimeKG graph-composition experiment for DDI link prediction. The experiment holds constant the DDI split, model architecture, candidate set, optimization procedure, relation-slot allocation, and evaluation protocol while varying only the support relations included in message passing.

Second, it provides a five-seed R-GCN evaluation. The main comparison is not based on a single favorable run; results are aggregated across seeds 42–46, and **G3** versus **G0** is additionally analyzed using paired seed-level differences.

Third, it extends the main experiment with a five-seed relation-level ablation. Seven relation types are examined individually, which makes it possible to test whether the descriptive performance of the fully heterogeneous graph can be attributed to a single relation family.

Fourth, it evaluates robustness and generalization beyond the main transductive benchmark through complementary classification, DDI-edge cold-start analysis, exploratory DDInter evaluation, and qualitative case analysis.

Fifth, it integrates the verified research artifacts into a web application that exposes ranking results, graph context, relation analysis, methodology information, grounded metadata, and independent evidence services. The application is designed to communicate methodological scope and limitations rather than to present research predictions as clinical advice.

1.6 Scope and Responsible Interpretation

CHEERS is an academic research prototype. The final experimental target is restricted to the PrimeKG `drug_drug` relation and therefore does not represent every possible clinical DDI concept. The learned decoder produces raw ranking scores. These scores are not calibrated probabilities and are not designed to estimate severity, contraindication strength, patient-specific risk, or treatment suitability.

The application’s graph context is also descriptive rather than causal. If two drugs share a gene, protein, or disease neighbor in the G3 graph, that shared entity demonstrates a stored graph association. It does not establish a biological mechanism for a predicted DDI. Similarly, FDA label excerpts and PubMed records are retrieved independently from the R-GCN. Their presence or absence does not validate or invalidate the model score.

1.7 Project Evolution

1.7.1 Initial Clinical-Inference Concept

The project began with a broader objective: a knowledge-graph-based clinical inference system intended to connect clinical features to diseases, treatment drugs, contraindications, and DDIs. The intended reasoning path was approximately

clinical feature $\rightarrow$ disease $\rightarrow$ treatment drug $\rightarrow$ interaction/contraindication check.

The interim system design therefore emphasized heterogeneous triple integration and multi-step inference rather than a single controlled link-prediction question.

1.7.2 Exploratory KGE Stage

During the interim stage, three knowledge-graph embedding models were explored independently: TransE [3], ComplEx [4], and RotatE [5]. These models were useful for understanding representation learning, filtered link prediction, and relation-specific behavior. However, their archived experiments were not aligned sufficiently for a scientifically fair final comparison: dataset snapshots, mappings, split sizes, and evaluation configurations differed across runs.

For this reason, their numerical results are treated as historical project evidence rather than final experimental findings.

1.7.3 Transition to a Controlled R-GCN Study

The research direction was narrowed so that the central experimental variable could be controlled explicitly. Rather than asking whether a large clinical graph could support a broad inference pipeline, the final project asks whether specific forms of biomedical graph context improve DDI link prediction under one standardized R-GCN framework.

This transition changed the project in four important ways. First, PrimeKG became the final model-training source. Second, the target task was fixed to canonical DDI link prediction. Third, graph variants were built from the same fixed DDI split. Fourth, the evaluation protocol was standardized around filtered ranking and matched-seed comparison.

The resulting final title, *Effect of Biomedical Knowledge Graph Composition on R-GCN-Based Drug–Drug Interaction Prediction*, reflects this narrower and more defensible research contribution.

2. Background and Related Work

2.1 Drug–Drug Interaction Prediction

A drug–drug interaction refers broadly to a situation in which the effects, exposure, efficacy, or safety profile of one drug are altered in the presence of another drug. Curated DDI resources are useful for clinical reference and pharmacovigilance, but they are necessarily incomplete and continuously evolving. Computational DDI prediction therefore commonly appears as a knowledge-base completion or link-prediction problem: given observed relationships, a model ranks candidate pairs that are not already represented as known positives.

This project does not attempt to replace curated DDI resources. Instead, it evaluates a graph-learning hypothesis about how representation context affects link ranking.

2.2 Biomedical Knowledge Graphs

A knowledge graph represents entities and their typed relations as triples

$$(h, r, t),$$

where h is a head entity, r is a relation type, and t is a tail entity. A biomedical knowledge graph may contain drugs, genes, proteins, diseases, pathways, phenotypes, and many other entity types connected by semantically different relations.

Heterogeneous biomedical graphs are attractive for DDI research because two drugs can be related indirectly through common molecular targets, metabolic enzymes, transporters, therapeutic indications, contraindications, or disease context. However, heterogeneous context can also add irrelevant edges, increase graph density, or introduce structural biases. The usefulness of a relation therefore has to be evaluated rather than assumed.

2.3 PrimeKG

PrimeKG is a precision-medicine knowledge graph that integrates multiple biomedical sources into a large heterogeneous network [1]. The published resource contains more than one hundred thousand nodes and millions of relationships spanning multiple biological scales. PrimeKG is particularly relevant to CHEERS because it includes drug–drug, drug–protein, and drug–disease relations in a common graph representation.

The final CHEERS experiment uses a selected subgraph of PrimeKG rather than the full multimodal graph. The project focuses on the DDI target relation and seven biomedical support relations:

- target,
- enzyme,
- transporter,
- carrier,
- indication,
- contraindication, and
- off-label use.

This selection supports a controlled comparison between DDI-only, molecular-context, disease-context, and combined heterogeneous graphs.

2.4 Existing DDI Resources and Related Prediction Methods

Existing DDI systems include both curated interaction resources and computational prediction methods. DrugBank and DDInter primarily provide established or curated drug and interaction information, whereas methods such as Decagon, SumGNN, and KnowDDI apply graph-based learning to DDI prediction [6, 7, 8, 9, 10].

CHEERS does not attempt to replace these resources or claim that knowledge-graph-based DDI prediction is itself novel. Instead, its main research focus is a controlled study of how biomedical graph composition affects DDI ranking when the target split, candidate set, R-GCN architecture, decoder, training procedure, and evaluation protocol are held fixed.

Accordingly, CHEERS is complementary to both curated DDI resources and previous graph-based prediction research. Relative to curated resources, it investigates model ranking of held-out or unobserved links rather than providing clinical interaction management. Relative to previous KG/GNN methods, its distinguishing emphasis is the controlled isolation of graph composition as the experimental variable, together with five-seed evaluation, relation-level ablation, cold-start analysis, external evaluation, and transparent deployment of the resulting research artifacts.

Table 1: Comparison with representative DDI resources and graph-based prediction methods.

System Method	/ Main Purpose	Difference from CHEERS
DrugBank [6]	Curated drug, pharmacological, target, and interaction information.	CHEERS is not a curated clinical drug database; it experimentally ranks held-out or unobserved DDI links and analyzes model behavior.
DDInter [7]	Curated DDI records and associated interaction information.	CHEERS uses DDInter as an exploratory external evaluation source rather than as an input to R-GCN training.
Decagon [8]	Graph-convolutional prediction of polypharmacy side-effect relationships.	Decagon develops a specialized graph model for side-effect prediction, whereas CHEERS keeps one R-GCN framework fixed and varies the biomedical relations supplied to it.
SumGNN [9]	Biomedical knowledge-graph subgraph summarization for multi-typed DDI prediction and interpretable reasoning.	CHEERS does not introduce a new subgraph architecture; it focuses on controlled G0-G3 graph-composition comparison and relation-level contribution analysis.
KnowDDI [10]	Interpretable DDI prediction using drug-pair-specific knowledge subgraphs.	CHEERS instead focuses on measuring the effect of heterogeneous graph composition under a fixed model and evaluation protocol.
CHEERS	Controlled R-GCN DDI prediction study with graph-composition, relation, cold-start, external-transfer, and application-level analyses.	Its main contribution is controlled analysis of graph composition and relation contribution rather than a fundamentally new GNN architecture.

2.5 Knowledge Graph Embedding Models

2.5.1 TransE

TransE models a relation as a translation in a continuous vector space [3]. For a true triple (h, r, t) , the learned embeddings are encouraged to satisfy

$$\mathbf{h} + \mathbf{r} \approx \mathbf{t}.$$

Its simplicity and scalability made it an appropriate early baseline during the exploratory phase of CHEERS.

2.5.2 ComplEx

ComplEx represents entities and relations in a complex-valued embedding space and uses a Hermitian bilinear product for scoring [4]. Its ability to model symmetric and antisymmetric patterns made it relevant to a project containing a dominant symmetric DDI relation.

2.5.3 RotatE

RotatE represents relations as rotations in complex space and was designed to model patterns including symmetry, antisymmetry, inversion, and composition [5]. It was explored during the interim project stage.

These three KGE models are retained in the final report only as project evolution and technical background. They are not the final controlled experiment.

2.6 Graph Neural Networks and Message Passing

Graph neural networks learn node representations by repeatedly aggregating information from neighboring nodes. A generic message-passing update can be written as

$$\mathbf{h}_i^{(\ell+1)} = \phi \left(\mathbf{h}_i^{(\ell)}, \text{AGG} \left\{ \psi \left(\mathbf{h}_j^{(\ell)}, e_{j,i} \right) : j \in \mathcal{N}(i) \right\} \right),$$

where $\mathcal{N}(i)$ is the neighborhood of node i , $e_{j,i}$ describes an edge, and ϕ and ψ are learnable or fixed transformations.

Standard graph convolution does not directly distinguish many relation types. Biomedical KGs require the model to treat, for example, an enzyme relation differently from an indication relation.

2.7 Relational Graph Convolutional Networks

R-GCN was introduced for multi-relational data and knowledge-base completion [2]. A common relation-aware update is

$$\mathbf{h}_i^{(\ell+1)} = \sigma \left(\mathbf{W}_0^{(\ell)} \mathbf{h}_i^{(\ell)} + \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} \mathbf{W}_r^{(\ell)} \mathbf{h}_j^{(\ell)} \right).$$

Here, $\mathcal{N}_i^r$ is the set of neighbors connected to node i by relation r , $\mathbf{W}_r$ is a relation-specific transformation, and $c_{i,r}$ is a normalization factor.

The final CHEERS implementation uses PyTorch Geometric’s `RGCNConv` layers. The key experimental idea is that the architecture remains fixed while the set of active relations in the input graph changes.

2.8 DistMult-Style Bilinear Decoding

DistMult is a bilinear knowledge-base embedding model in which each relation is represented by a diagonal interaction vector [11]. CHEERS uses a symmetric DistMult-style DDI decoder after the R-GCN encoder:

$$s(u, v) = \sum_{k=1}^d z_{u,k} r_k z_{v,k},$$

where $\mathbf{z}_u$ and $\mathbf{z}_v$ are the final encoded node representations and $\mathbf{r}$ is the learned DDI relation vector.

Because scalar multiplication is commutative,

$$z_{u,k} r_k z_{v,k} = z_{v,k} r_k z_{u,k},$$

and therefore

$$s(u, v) = s(v, u).$$

This symmetry is compatible with CHEERS’s canonical treatment of DDI pairs as undirected pairs for splitting and evaluation.

2.9 Link Prediction

Link prediction estimates the plausibility of graph edges that are not observed in the training graph. In knowledge-graph evaluation, a held-out positive target is ranked against a set of candidate entities. The resulting rank is then used to compute metrics such as MRR and Hits@K.

A major methodological issue is the distinction between *unobserved* and *negative*. Biomedical graphs are incomplete, so a drug pair not present as a known positive cannot automatically be treated as a confirmed non-interaction. CHEERS therefore uses the term *sampled unobserved pair* for sampled negative-training/classification examples.

2.10 Transductive and Inductive Evaluation

In a transductive graph-learning experiment, nodes appearing in validation or test relationships may already be known during training, although the held-out target edges themselves are removed. This is the setting of the main CHEERS G0–G3 experiment.

A fully inductive unseen-drug experiment would require the model to represent a drug not assigned a learned node-ID embedding during training, typically using transferable molecular, textual, or structural features. CHEERS does not implement this setting. Its secondary cold-start experiment instead removes DDI training/message-passing edges for selected known drugs.

2.11 Relation-Level Ablation

Ablation analysis removes or isolates components of a system to study their contribution. CHEERS uses an *add-one* relation-ablation design. Starting from the DDI-only **G0** graph, each ablation variant adds one biomedical relation and its reverse message-passing relation. This permits direct paired comparison with the same DDI backbone.

2.12 Cold-Start Evaluation

Cold start generally refers to prediction under limited historical information. In recommender systems, a new user or item may lack interactions; in graph learning, a new or sparsely connected entity may lack observed edges. CHEERS implements a narrower DDI-edge cold-start setting: selected drug nodes remain present, but their DDI training/message-passing edges are removed. In **G3**, those nodes may still receive messages through non-DDI biomedical relations.

2.13 Evaluation Metrics

2.13.1 Mean Reciprocal Rank

For Q directional ranking queries,

$$\text{MRR} = \frac{1}{Q} \sum_{q=1}^Q \frac{1}{\text{rank}_q}.$$

MRR rewards models that place the true target near the top of the ranked candidate list.

2.13.2 Hits@K

Hits@K is

$$\text{Hits@K} = \frac{1}{Q} \sum_{q=1}^Q \mathbf{1}[\text{rank}_q \leq K].$$

The authoritative final experiment reports Hits@1, Hits@5, and Hits@10.

2.13.3 Classification Metrics

For complementary binary evaluation,

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

$$\text{F1} = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}},$$

and

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Classification is secondary in CHEERS because the primary scientific task is candidate ranking, and the sampled-unobserved negatives are not clinically verified non-interactions.

2.14 Paired Statistical Analysis Across Seeds

If two graph variants are trained with the same set of random seeds, seed-wise results can be paired. For metric m and seed s , CHEERS defines

$$\Delta_s = m(\mathbf{G3}, s) - m(\mathbf{G0}, s).$$

The mean paired effect is

$$\bar{\Delta} = \frac{1}{n} \sum_{s=1}^n \Delta_s.$$

The final validation reports pointwise 95% paired t intervals and exact two-sided sign-flip tests over the five matched seed differences. Because the sample size is small and all runs use one fixed split, these intervals describe training-seed variability only; they do not quantify dataset, split, or clinical uncertainty.

3. Dataset and Knowledge Graph Construction

3.1 Final Data-Source Scope

The final model-training experiment is based on PrimeKG. Other resources appear in CHEERS for external evaluation or application-level metadata, but they are not inputs to the final R-GCN training graph.

DDInter is an open DDI resource providing curated interaction records and associated annotations [7]. CHEERS uses it only for exploratory external evaluation, not for model training.

Table 2: Role of principal data sources in the final CHEERS project

Source	Purpose	R-GCN train- ing	Scope
PrimeKG	DDI target and biomedical support graph	Yes	Final methodology
DDInter	External DDI ranking pilot	No	External evaluation
NCBI Gene	Gene/protein metadata	No	Web metadata
MONDO	Disease map- ping/description	No	Web metadata
openFDA/FDA labels	Independent label evidence	No	Web evidence
PubMed	Independent literature re- trieval	No	Web evidence
OCR/Tesseract	Label-text recognition	No	Web input utility
Disease/nutrition records	Educational disease informa- tion	No	Auxiliary web feature

3.2 PrimeKG DDI Extraction

The PrimeKG `drug_drug` rows are symmetric in the source graph: each undirected DDI pair appears in two orientations. The processed project records contain:

- 2,672,628 directed `drug_drug` rows;
- 1,336,314 unique undirected DDI pairs;
- 1,336,314 reverse duplicates;
- 0 self-loops.

For splitting, CHEERS canonicalizes each pair by imposing a deterministic ordering on the two drug endpoints and retaining one canonical undirected representation. This step is necessary to prevent one orientation of a drug pair from entering training while the reverse orientation enters validation or test.

3.3 Canonical DDI Split

The 1,336,314 canonical pairs are divided into one fixed train/validation/test split.

Table 3: Canonical DDI split used by all final graph variants

Split	Canonical DDI pairs	Percentage
Training	1,069,080	80.00%
Validation	133,620	10.00%
Test	133,614	10.00%
Total	1,336,314	100.00%

The final graph population contains 13,094 shared nodes, including 4,278 candidate drug nodes used for DDI ranking.

3.4 Leakage Safeguards

All graph variants use the same canonical split. Validation and test DDI pairs are excluded from the message-passing DDI adjacency. Consequently, the R-GCN cannot propagate information directly through the target DDI edges being evaluated.

A complete known-positive mask is retained for two purposes:

1. to prevent known positives from being sampled as unobserved negative examples; and
2. to filter other known-positive candidate drugs during ranking.

The existence of the full known-positive mask does not imply that validation/test target edges are used for training messages. It is an evaluation and sampling safeguard.

3.5 Transductive Setting

The main split is edge-based rather than entity-disjoint. Drugs in validation and test therefore remain known nodes and retain learned node embeddings. This makes the primary G0-G3 experiment transductive.

This setting is appropriate for the project’s main research question, which concerns graph composition under a controlled fixed node population. It does not answer the stronger question of how the system would represent an entirely unseen drug.

3.6 Biomedical Support Relations

CHEERS selects seven PrimeKG support relations. The forward-edge counts are shown below.

Table 4: Forward biomedical support-edge counts

Relation	Forward edges	Directed edges after reverse addition
Target	16,380	32,760
Enzyme	5,317	10,634
Transporter	3,092	6,184
Carrier	864	1,728
Indication	9,388	18,776
Contraindication	30,675	61,350
Off-label use	2,568	5,136

Drug-Gene/Protein relations are Target, Enzyme, Transporter, and Carrier. Drug-Disease relations are Indication, Contraindication, and Off-label use.

3.7 Graph Variants

Four graph variants are constructed from the same training DDI backbone.

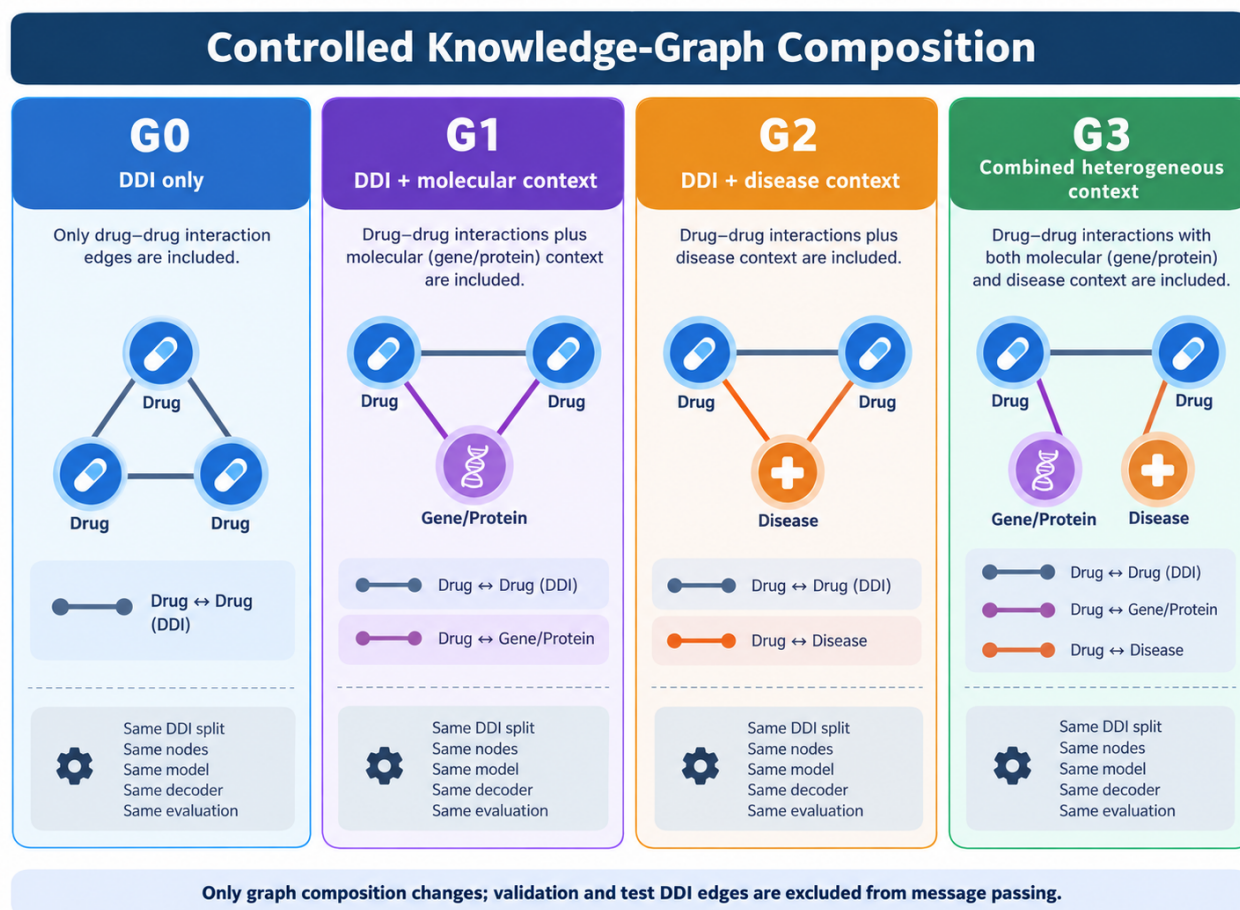


Figure 1: Controlled PrimeKG-derived graph-composition variants used in the final R-GCN experiment. G0 contains only drug-drug interaction edges; G1 adds Drug-Gene/Protein molecular context; G2 adds Drug-Disease context; and G3 combines both biomedical context families. The DDI split, node population, model architecture, decoder, and evaluation protocol are kept fixed across all variants.

3.7.1 G0: DDI Only

G0 contains only the directed representation of the 1,069,080 canonical training DDI pairs. Because each canonical DDI is represented in both directions for message passing, the graph contains 2,138,160 directed edges and one active relation.

3.7.2 G1: DDI + Drug-Gene/Protein

G1 augments **G0** with Target, Enzyme, Transporter, and Carrier edges plus their reverse message-passing relations. It contains 2,189,466 directed edges and nine active relations.

3.7.3 G2: DDI + Drug-Disease

G2 augments **G0** with Indication, Contraindication, and Off-label use edges plus reverse relations. It contains 2,223,422 directed edges and seven active relations.

3.7.4 G3: Combined Heterogeneous Graph

G3 contains the DDI backbone plus all seven biomedical support relations and their reverse edges. It contains 2,274,728 directed edges and all 15 allocated relation slots are active.

Table 5: Final graph variants

Graph	Composition	Directed edges	Active relations
G0	DDI only	2,138,160	1
G1	DDI + Drug-Gene/Protein	2,189,466	9
G2	DDI + Drug-Disease	2,223,422	7
G3	DDI + Drug-Gene/Protein + Drug-Disease	2,274,728	15

3.8 Global Relation Mapping

All graph variants allocate the same 15 relation slots so that relation-parameter allocation is not itself a confound.

Inactive relation weights in a graph variant receive no messages. The DDI target relation remains one symmetric relation ID rather than being assigned an independent reverse-relation ID.

Table 6: Global relation-ID mapping

ID	Relation
0	drug_drug
1	target
2	reverse target
3	enzyme
4	reverse enzyme
5	transporter
6	reverse transporter
7	carrier
8	reverse carrier
9	indication
10	reverse indication
11	contraindication
12	reverse contraindication
13	off-label use
14	reverse off-label use

3.9 Purpose of Reverse Support Relations

The semantic source relations are directional, but message passing benefits from allowing information to flow in both directions. For example, a Drug $\rightarrow$ Gene target edge enables the drug to receive information from the gene only if the adjacency used by the layer contains the required incoming orientation. Adding reverse support edges therefore enables bidirectional propagation while retaining explicit relation identity.

3.10 Controlled Experimental Variable

The graph composition is the principal independent variable. The following are controlled across G0–G3:

- canonical DDI train/validation/test split;
- node population;
- 4,278 candidate drugs;
- allocated relation slots;
- R-GCN architecture;
- DDI decoder;
- optimizer and learning-rate configuration;
- positive/negative sampling ratio;
- checkpoint-selection rule; and
- filtered-ranking protocol.

This design permits observed performance differences to be interpreted primarily as graph-composition effects, subject to optimization noise and finite-seed uncertainty.

4. R-GCN Methodology

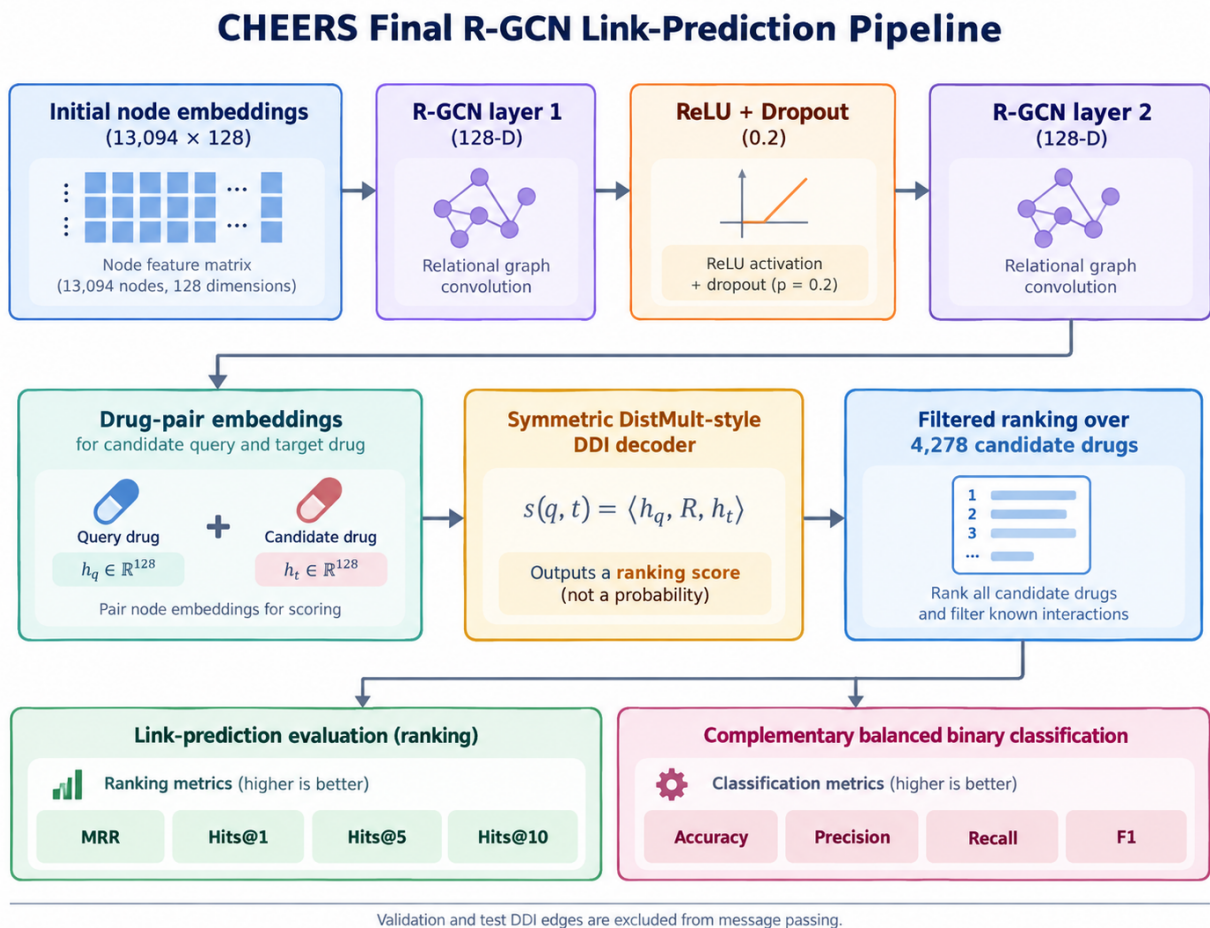


Figure 2: Final CHEERS R-GCN link-prediction pipeline. Each node begins with a learned 128-dimensional embedding, followed by two relational graph-convolution layers with ReLU and dropout between them. Drug-pair representations are scored using a symmetric DistMult-style DDI decoder. The output is a ranking score, not a calibrated probability or clinical-risk value.

4.1 Overall Model Pipeline

For each graph variant, CHEERS trains the same two-layer R-GCN encoder and symmetric DDI decoder. The computational flow is

$$\begin{aligned} \text{graph variant} &\rightarrow \text{node embeddings} \rightarrow \text{R-GCN layer 1} \\ &\rightarrow \text{ReLU + dropout} \rightarrow \text{R-GCN layer 2} \rightarrow \text{DDI score.} \end{aligned}$$

The graph determines which typed neighbor messages are available. The model architecture itself is unchanged.

4.2 Learned Node Embeddings

The shared graph contains 13,094 nodes. Each node is assigned a learned 128-dimensional input embedding. The initial embedding matrix therefore has shape

$$13,094 \times 128.$$

Because these are node-ID embeddings, the model is transductive. A completely unseen entity has no trained row in this embedding matrix.

4.3 Relational Graph Convolution

The final implementation uses two PyTorch Geometric `RGCNConv` layers. Relation-specific messages propagate over the graph. The first layer is followed by ReLU and dropout; the second produces the final encoded representation $\mathbf{z}_i$ for each node.

No basis-decomposition setting is explicitly configured in the verified final implementation. The model therefore uses the relation handling supplied by the instantiated `RGCNConv` configuration.

4.4 Activation and Regularization

After the first R-GCN layer, CHEERS applies

$$\text{ReLU}(x) = \max(0, x)$$

followed by dropout with rate 0.2. Dropout reduces co-adaptation of hidden features during training. The second layer directly produces the embeddings used by the DDI decoder.

4.5 Symmetric DDI Decoder

For drug nodes u and v , the model score is

$$s(u, v) = \sum_{k=1}^{128} z_{u,k} r_k z_{v,k}.$$

The learned vector $\mathbf{r} \in \mathbb{R}^{128}$ parameterizes the DDI relation. The decoder is symmetric:

$$s(u, v) = s(v, u).$$

The score is an unconstrained real-valued ranking quantity. No sigmoid-calibration procedure is used to convert it to a clinical probability.

4.6 Training Objective

Training uses binary cross-entropy with logits over positive training DDI pairs and sampled unobserved pairs. If $y \in \{0, 1\}$ and s is a logit,

$$\mathcal{L}(s, y) = -y \log \sigma(s) - (1 - y) \log(1 - \sigma(s)),$$

where

$$\sigma(s) = \frac{1}{1 + e^{-s}}.$$

Although BCE uses the logistic transformation internally, this does not make the exported raw ranking score a calibrated probability.

4.7 Negative Sampling

Training samples 100,000 positive DDI pairs per epoch and uses a 1:1 positive-to-unobserved ratio. Candidate unobserved pairs must satisfy:

- both endpoints are candidate drugs;
- the endpoints are not identical;
- the pair is not in the complete known-positive DDI set.

The term *negative* is used operationally by the loss function, but scientifically the samples are described as *sampled unobserved pairs*. A missing graph edge can reflect an unknown, uncured, unsupported, or genuinely absent relationship.

4.8 Optimization and Hyperparameters

Table 7: Final R-GCN training configuration

Parameter	Value
Input embedding dimension	128
Hidden dimension	128
R-GCN layers	2
Dropout	0.2
Decoder	Symmetric DistMult-style bilinear decoder
Optimizer	Adam
Learning rate	0.001
Weight decay	10^{-5}
Maximum epochs	500
Early-stopping patience	10
Positive samples per epoch	100,000
Negative ratio	1:1
Training seeds	42, 43, 44, 45, 46
Checkpoint criterion	Validation BCE
Allocated relation slots	15
Model parameters	2,200,704

4.9 Model Selection

Each graph/seed run is trained independently. The best checkpoint is selected using validation binary cross-entropy. Test data are not used to choose the checkpoint.

The web application uses the preserved G3 seed-44 checkpoint. This checkpoint is used because it is the verified/exported model associated with the final lightweight runtime, not because seed 44 is assumed to represent the expected mean performance.

4.10 Training Environment

The documented university training environment used:

- Python 3.10.20;
- PyTorch 2.2.1 with CUDA 12.1;
- PyTorch Geometric 2.5.3;
- CUDA-enabled execution;
- four NVIDIA GeForce RTX 2080 Ti GPUs available on the server.

Final jobs were isolated to a selected physical GPU using `CUDA_VISIBLE_DEVICES`. The local web runtime is intentionally lighter and does not require the full training environment.

4.11 Filtered Ranking Evaluation

4.11.1 Directional Queries

The test set contains 133,614 canonical undirected DDI pairs. Each pair is evaluated in both directions, producing

$$2 \times 133,614 = 267,228$$

directional ranking queries.

4.11.2 Candidate Set

For each query, the model scores all 4,278 candidate drugs.

4.11.3 Filtering

Before computing the rank of the held-out target:

1. other known-positive DDI candidates are filtered;
2. the source drug itself is filtered;
3. the current held-out target is restored even though it belongs to the complete known-positive set.

This prevents a model from being penalized for ranking another already-known positive above the current test target.

4.11.4 Primary Metrics

The primary metrics are MRR, Hits@1, Hits@5, and Hits@10. Hits@3 is not included in the authoritative final G0–G3 result summary and is therefore omitted from the main thesis result tables.

4.12 Complementary Classification Evaluation

The project also evaluates the trained checkpoints as binary discriminators without retraining. Validation data are used to choose a score threshold that maximizes F1 independently for each graph/seed. That threshold is then frozen and applied to the held-out test set.

The test classification set contains 133,614 positive pairs and an equal number of fixed sampled-unobserved pairs. Because thresholded classification answers a different question from full candidate ranking, it is reported as complementary analysis rather than the primary endpoint.

5. Main Experimental Results

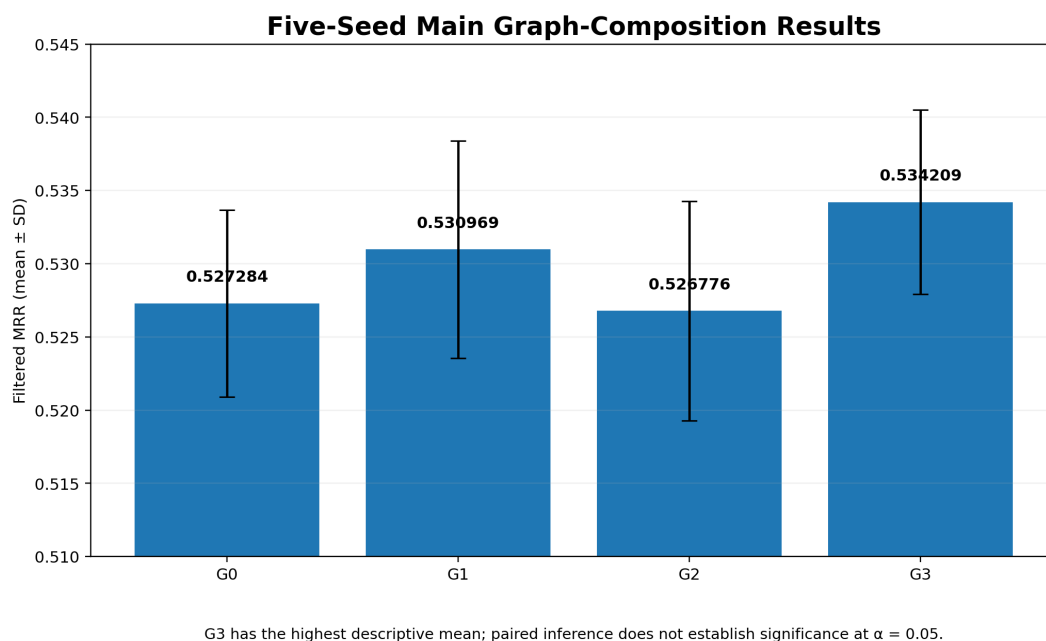


Figure 3: Five-seed MRR comparison across G0–G3. Error bars show sample standard deviation across seeds.

5.1 Five-Seed Graph-Composition Comparison

The main experiment compares **G0–G3** over seeds 42–46. Table 8 reports the authoritative mean and sample standard deviation.

The ranking order by mean MRR is $G3 > G1 > G0 > G2$. The central descriptive observation is therefore that the combined heterogeneous graph achieves the highest average ranking performance, whereas adding disease relations alone does not improve the DDI-only baseline.

Table 8: Five-seed filtered-ranking results

Graph	MRR	Hits@1	Hits@5	Hits@10
G3	0.534209 $\pm$ 0.006288	0.486290 $\pm$ 0.005899	0.580468 $\pm$ 0.007000	0.618074 $\pm$ 0.007312
G1	0.530969 $\pm$ 0.007414	0.484194 $\pm$ 0.007010	0.575418 $\pm$ 0.007907	0.612683 $\pm$ 0.007986
G0	0.527284 $\pm$ 0.006373	0.480464 $\pm$ 0.005801	0.572694 $\pm$ 0.007746	0.609447 $\pm$ 0.009457
G2	0.526776 $\pm$ 0.007482	0.480197 $\pm$ 0.006691	0.571716 $\pm$ 0.008899	0.608608 $\pm$ 0.009758

5.2 Matched G0 and G3 Seed Results

The final paired-validation artifact preserves complete G0/G3 per-seed metrics.

Table 9: Matched G0 and G3 MRR across seeds

Seed	G0 MRR	G3 MRR	Δ MRR
42	0.532557	0.537475	+0.004918
43	0.534970	0.538355	+0.003385
44	0.519827	0.540359	+0.020532
45	0.526091	0.527911	+0.001820
46	0.522979	0.526943	+0.003964

G3 achieves a positive MRR difference in every matched seed. However, the magnitude is heterogeneous: seed 44 contributes a notably larger difference than the other four seeds. This motivates paired uncertainty analysis rather than interpretation of the mean difference alone.

5.3 Complementary Classification Results

Table 10: Five-seed complementary classification performance

Graph	Accuracy	Precision	Recall	F1
G0	0.919181 $\pm$ 0.002974	0.935422 $\pm$ 0.003051	0.900533 $\pm$ 0.003844	0.917643 $\pm$ 0.003075
G1	0.921896 $\pm$ 0.002155	0.936814 $\pm$ 0.002077	0.904826 $\pm$ 0.004422	0.920536 $\pm$ 0.002335
G2	0.919909 $\pm$ 0.001713	0.935307 $\pm$ 0.002079	0.902233 $\pm$ 0.004752	0.918463 $\pm$ 0.001958
G3	0.922718 $\pm$ 0.003036	0.937268 $\pm$ 0.001843	0.906082 $\pm$ 0.005880	0.921403 $\pm$ 0.003300

Classification and ranking are related but not equivalent objectives. Ranking asks whether a true target is placed near the top among thousands of candidates; thresholded classification asks whether selected positives can be separated from sampled unobserved pairs at a chosen decision threshold. CHEERS therefore uses the ranking results as the principal scientific endpoint.

5.4 Paired Statistical Validation

For each metric, the final analysis computes G3 minus G0 within each matched seed.

Table 11: Paired G3–G0 statistical validation

Metric	Mean paired difference	95% paired t interval	Exact sign-flip p
MRR	+0.006924	[-0.002624, +0.016473]	0.0625
Hits@1	+0.005826	[-0.002127, +0.013779]	0.0625
Hits@5	+0.007775	[-0.003753, +0.019302]	0.0625
Hits@10	+0.008626	[-0.004399, +0.021652]	0.0625

All four mean differences favor G3 descriptively, but every 95% interval includes zero. The exact two-sided sign-flip test also does not cross the conventional 0.05 threshold. After Holm adjustment across the four reported metrics, the adjusted p -value is 0.25 for each comparison.

5.5 Interpretation of “Interval Includes Zero”

A paired confidence interval that includes zero means that the observed seed-to-seed variation is compatible with an average difference of zero at the stated confidence level. It does not prove that the two models are equivalent, and it does not prove that graph composition has no effect. Instead, it limits the strength of the conclusion that can be drawn from five seeds on one fixed split.

The final CHEERS conclusion is therefore deliberately conservative:

The fully heterogeneous G3 graph shows a small and consistent descriptive advantage over the DDI-only G0 baseline, but the current five-seed fixed-split experiment does not establish that advantage as statistically reliable at the conventional 0.05 level.

5.6 Why G2 Matters

G2 is important because it demonstrates that adding biomedical context is not automatically beneficial. The Drug–Disease graph has more support edges than G1, yet its mean MRR is slightly below G0. This result argues against a simple “more edges are better” interpretation and motivates relation-level analysis.

6. Relation-Level Ablation

6.1 Motivation

The main experiment compares groups of relations. Because G3 combines seven biomedical relation types, its descriptive improvement cannot by itself reveal which relation is responsible. CHEERS therefore performs a single-relation add-one study.

6.2 Ablation Construction

Every ablation begins with the same G0 DDI backbone and adds one forward biomedical relation plus its reverse message-passing relation:

- A1: Target,
- A2: Enzyme,
- A3: Transporter,
- A4: Carrier,
- A5: Indication,
- A6: Contraindication,

- A7: Off-label use.

The current versioned ranking study uses seeds 42–46 and paired G0 baselines for the same seeds.

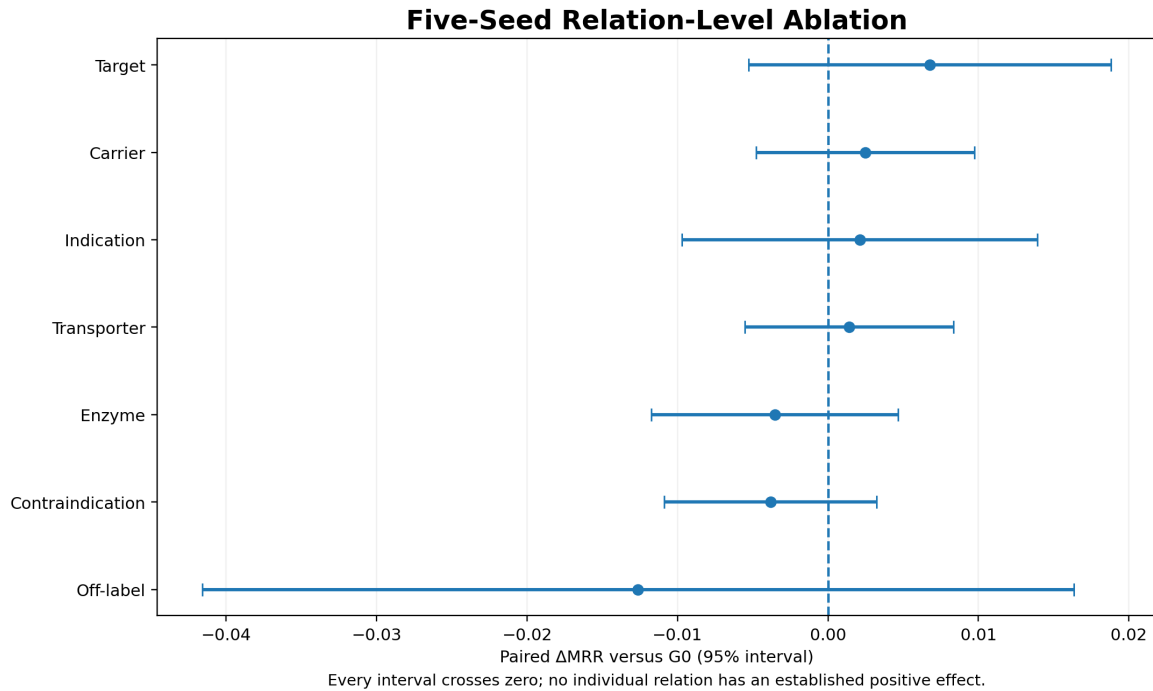


Figure 4: Five-seed paired relation-ablation MRR effects relative to G0. Every pointwise 95% paired interval crosses zero.

6.3 Five-Seed Ablation Results

Table 12: Five-seed relation-ablation ranking results

Relation	Forward edges	Mean MRR	Mean Δ MRR	Wins	95% interval
Target	16,380	0.534051	+0.006766	4/5	[-0.005290, 0.018823]
Carrier	864	0.529763	+0.002479	3/5	[-0.004763, 0.009721]
Indication	9,388	0.529391	+0.002107	2/5	[-0.009703, 0.013916]
Transporter	3,092	0.528694	+0.001410	4/5	[-0.005538, 0.008357]
Enzyme	5,317	0.523753	-0.003531	1/5	[-0.011735, 0.004672]
Contraindication	30,675	0.523467	-0.003817	1/5	[-0.010881, 0.003247]
Off-label use	2,568	0.514659	-0.012625	1/5	[-0.041582, 0.016332]

Target has the largest descriptive mean improvement and wins four of five paired runs. Carrier and Transporter also have positive mean effects, despite relatively small edge counts. Conversely, Contraindication is the largest relation by forward edge count but has a negative mean effect.

6.4 Uncertainty

Every relation-level 95% paired interval includes zero. Consequently, none of the individual relations can be described as having an established improvement over G0.

This is a particularly important result because the three-seed exploratory analysis produced a different descriptive ordering. With seeds 42–44, Indication appeared strongest by mean MRR, and Carrier/Transporter had strong win patterns. Adding seeds 45 and 46 changed the ranking, with Target becoming the strongest mean relation. The change illustrates why small-seed exploratory conclusions should not be treated as stable findings.

6.5 Relation Size Does Not Determine Contribution

The ablation results do not show a monotonic relationship between edge count and predictive improvement. Carrier is the smallest relation but has a positive mean paired effect, while Contraindication is one of the largest relations and has a negative mean effect. Relation semantics, graph topology, degree distribution, and optimization dynamics may all matter.

CHEERS does not claim a causal explanation for these differences. Establishing mechanism would require additional graph-structural analysis, multiple data splits, and potentially alternative GNN architectures.

6.6 Ranking Versus Classification

The historical relation-level classification analysis is based on three seeds and is therefore retained as secondary evidence rather than merged into the current five-seed ranking release. It showed that a relation’s effect on ranking need not match its effect on thresholded F1.

This distinction reinforces a broader methodological point: “useful relation” is not an evaluation-independent property. A relation can help move true targets upward in a long candidate list without improving separation at a fixed binary threshold.

6.7 Ablation Conclusion

The five-seed ablation supports three conclusions:

1. no individual relation has an established positive effect under the current fixed-split protocol;
2. Target has the strongest descriptive mean effect;
3. edge count alone does not explain observed relation usefulness.

The G3 result should therefore be interpreted as a property of the combined heterogeneous graph rather than attributed confidently to one relation.

7. Generalization and External Evaluation

7.1 Purpose

The main experiment is transductive and preserves the complete node population. CHEERS adds secondary analyses to examine whether conclusions remain meaningful under reduced DDI supervision and when ranking pairs from an external DDI resource.

7.2 DDI-Edge Cold-Start Definition

The implemented cold-start experiment removes DDI training/message-passing edges associated with selected “cold” drug nodes. The nodes themselves remain in the graph and retain learned node-ID embeddings. In G3, they may additionally retain non-DDI biomedical support relations. This protocol should not be described as fully inductive unseen-drug evaluation.

7.3 Cold-Drug Cohort

The cohort is selected once using split seed 42 and then frozen across model seeds 42–44. Eligibility requires:

- total known DDI degree of at least 20;
- G3 biomedical graph degree of at least 5;
- existing test DDI degree of at least 10.

A total of 1,969 drugs satisfy the criteria. A deterministic 10% stratified sample over four DDI-degree strata yields 196 cold drugs, with 49 selected from each stratum.

The cohort design intentionally ensures that selected drugs have non-DDI context available in G3. It also uses existing test-degree information and is therefore not a fully test-blind cohort-construction procedure.

7.4 Cold-to-Warm Results

For 17,176 cold-to-warm directional queries, the reproduced results are:

Table 13: Cold-to-warm filtered-ranking results

Graph	MRR	Hits@1	Hits@5	Hits@10
G0	0.140603 ± 0.058159	0.135130	0.142932	0.146483
G3	0.010184 ± 0.007833	0.007006	0.009975	0.011896

G0 performs substantially better in this condition. The result indicates that heterogeneous context does not automatically compensate for removed DDI connectivity.

7.5 Cold-to-Cold Results

For 651 cold-cold pairs, corresponding to 1,302 directional queries:

Table 14: Cold-to-cold filtered-ranking results

Graph	MRR	Hits@1	Hits@5	Hits@10
G0	0.000990 ± 0.000177	0.000000	0.000000	0.000000
G3	0.008430 ± 0.004076	0.001280	0.005376	0.011009

G3 is better in this more severe condition, although absolute performance remains low. The result suggests that retained biomedical connectivity can provide some ranking signal when both endpoints lack DDI training edges.

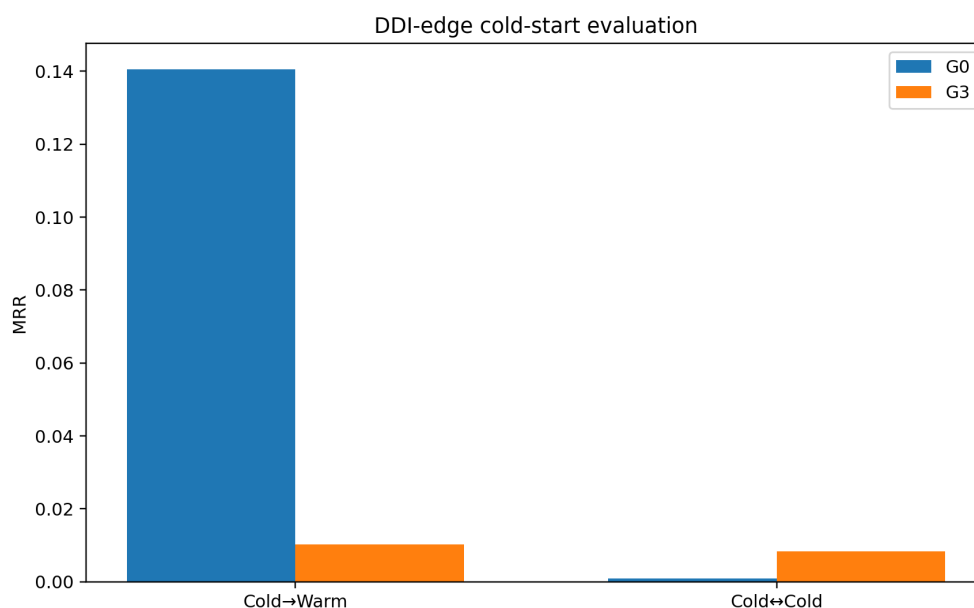


Figure 5: DDI-edge cold-start MRR comparison. The two evaluation conditions produce different conclusions.

7.6 Cold-Start Interpretation

The two cold-start conditions point in different directions. It would therefore be incorrect to summarize the experiment as “G3 solves cold start.” A more accurate conclusion is:

The usefulness of heterogeneous context depends on the cold-start condition. G0 is stronger for cold-to-warm ranking, while G3 retains more signal for cold-to-cold ranking. The current experiment remains transductive at the node level and does not test unseen-drug induction.

7.7 DDInter External Evaluation

DDInter provides an external DDI resource [7]. CHEERS uses conservative exact normalized-name mapping rather than fuzzy matching.

The preparation identified:

- 222,383 physical rows;
- 160,235 unique symmetric pairs;
- 1,939 drugs;
- 1,614 exact normalized-name mappings;
- 325 unmapped drugs;
- 49,105 mapped pairs absent from the complete PrimeKG positive mask;
- 98,210 directional evaluation queries.

7.8 External Ranking Results

The external pilot evaluates seed 44 only.

Table 15: Exploratory DDInter ranking results, seed 44

Model	MRR	Hits@1	Hits@5	Hits@10
G0	0.012544	0.003503	0.012697	0.022421
G3	0.012088	0.003442	0.012249	0.021383
Structural-only diagnostic	0.016503	0.004786	0.017340	0.030089
Hybrid diagnostic	0.012446	0.003574	0.012585	0.022330

G3 does not outperform G0 in this pilot. This is useful negative evidence: the descriptive advantage seen on the internal fixed PrimeKG split does not automatically transfer to an external DDI source.

7.9 Limitations of External Evaluation

The external analysis is not a clinical validation study. It is limited by:

- one selected model seed;
- exact-name mapping and unmapped drugs;
- differences in source curation and DDI semantics;
- exclusion of pairs already represented as PrimeKG positives;
- possible identity/formulation differences across resources.

7.10 Qualitative Case Studies

The verified three-pair set includes:

- Colchicine–Probenecid;
- Acetylsalicylic acid–Ibuprofen;
- Metformin–Glipizide.

All are neutral examples under the preserved best-rank categorization. For Colchicine–Probenecid, the G3 context contains 33 shared entities and includes gene/protein context such as ALB, CYP2C8, and CYP3A4. These shared neighbors are descriptive graph associations only. They are not presented as causal mechanisms.

Independent FDA and PubMed evidence may be shown by the web application, but it is retrieved separately and does not participate in the ranking score.

8. CHEERS System Architecture and Implementation

8.1 Purpose of the Research Application

The CHEERS web application makes the final research artifacts inspectable without requiring users to run the original PyTorch/PyG training environment. Its purpose is to expose the model’s ranking output, the G3 graph context, experimental summaries, and independent evidence in a transparent research interface.

The application is not intended to provide medication advice or clinical decision support.

8.2 Architecture Overview

The application contains three principal layers:

1. a React/Vite frontend;
2. a FastAPI backend; and
3. frozen research/runtime artifacts, including the lightweight NumPy predictor and G3 context stores.

Independent evidence clients query FDA/openFDA and PubMed when network access is available. Browser-side OCR uses local Tesseract.js assets.

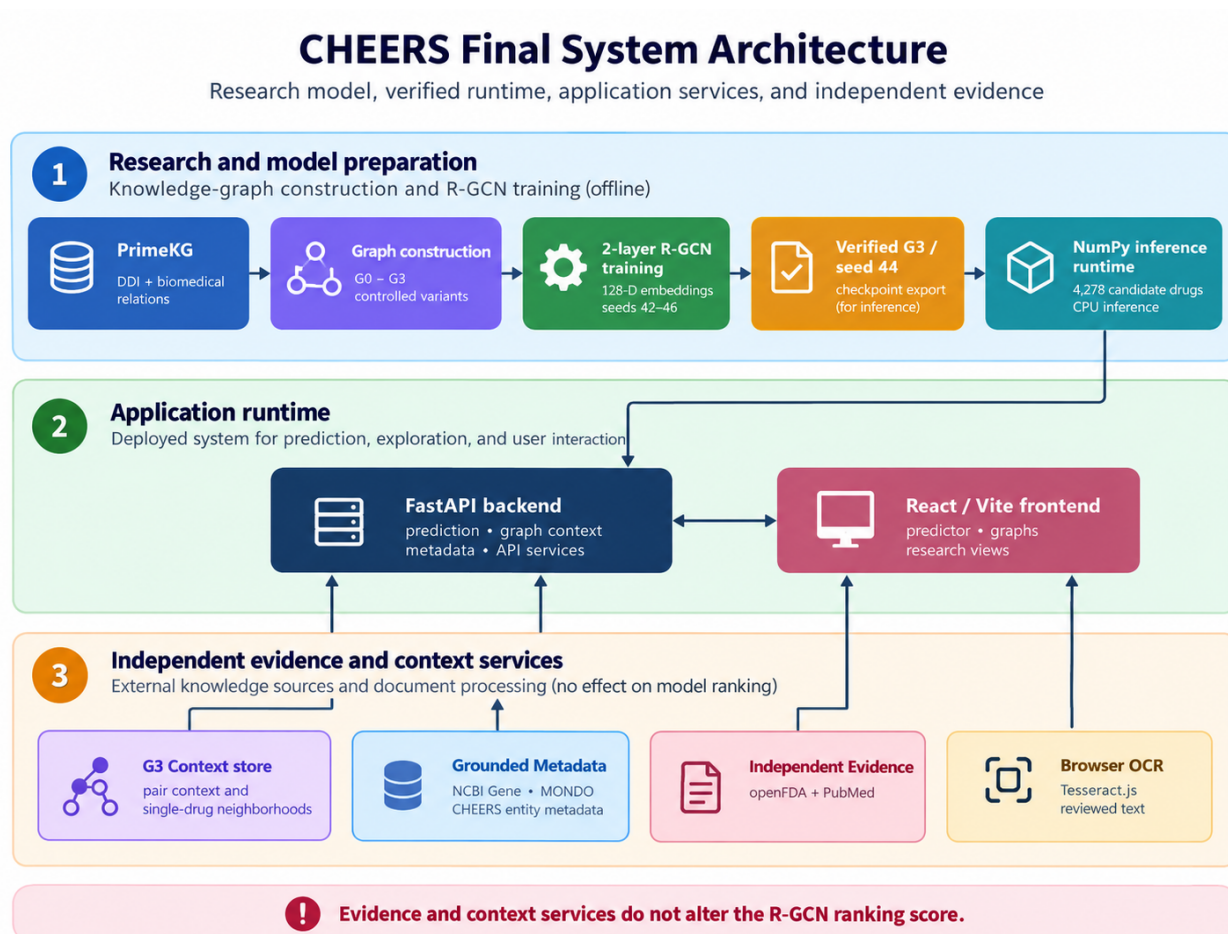


Figure 6: Final CHEERS system architecture. The offline research pipeline constructs PrimeKG-derived graph variants, trains the R-GCN models, and exports the verified G3 seed-44 checkpoint for lightweight NumPy inference. The deployed application separates prediction, graph-context services, grounded metadata, independent FDA/PubMed evidence retrieval, and browser-side OCR. Evidence and context services support interpretation and navigation, but they do not alter the R-GCN ranking score.

8.3 Selected Demonstration Model

The deployed research demonstration uses:

- graph variant G3;
- seed 44;
- checkpoint `checkpoints/rgcn_multiseed/G3_seed44_best.pt`;
- best epoch 499.

G3 is the highest-mean graph in the five-seed internal experiment. Seed 44 is used because it is the preserved and fully verified/exported checkpoint, not because it is assumed to be the statistically representative seed.

8.4 Lightweight NumPy Runtime

The application does not execute the full R-GCN encoder for every user query. Instead, the selected checkpoint is exported into a lightweight runtime containing the already-computed candidate embeddings, the learned DDI relation vector, drug metadata, and a packed known-positive mask.

For a selected drug u , candidate v is scored as

$$s(u, v) = \sum_k z_{u,k} r_k z_{v,k}$$

using NumPy. Known-positive and self candidates are then removed from the Top-K output.

This design removes the runtime dependency on PyTorch, PyTorch Geometric, CUDA, and the university training server.

8.5 Runtime Verification

The lightweight export is validated against the original model artifacts. The final verification includes exact Top-K reproduction for a fixed drug query and a full-test numerical tolerance on the order of 10^{-6} . The preserved G3 seed-44 full-test MRR is 0.540359.

The runtime should therefore be described as an exported implementation of the same decoder over frozen R-GCN embeddings, not as a separate approximate model.

8.6 FastAPI Backend

The FastAPI service loads the model runtime, G3 graph context, training-DDI neighborhood data, entity metadata, disease information, FDA-label evidence client, and PubMed literature client during application startup.

Important API routes include:

Table 16: Principal CHEERS API endpoints

Method	Path	Purpose
GET	/api/health	Runtime and component health
GET	/api/model	Model/deployment metadata
GET	/api/experiment	Five-seed graph-composition results
GET	/api/classification	Classification results
GET	/api/relation-analysis	Five-seed ablation and historical analysis
GET	/api/verification	Verification summary
GET	/api/drugs/search	Drug search/autocomplete
POST	/api/drugs/match-text	Conservative matching from reviewed OCR text
POST	/api/predict	Top-K unobserved DDI ranking
GET	/api/context/pair	Pair-specific G3 graph context
GET	/api/context/drug	Paginated one-hop drug neighborhood
GET	/api/evidence/pair	Independent FDA/PubMed evidence
GET	/api/public/medicine	Grounded medicine information
GET	/api/public/disease	Grounded disease information

8.7 React/Vite Frontend

The final React/Vite application includes public information pages and research-focused views. Hash routing and lazy-loaded pages are used to organize the interface. The principal research views include:

- DDI Predictor;
- Pair Graph Explorer;
- Subgraph Explorer;
- Evidence view;
- Experiment results;
- Relation-ablation results;
- methodology and scope explanations.

8.8 Public Home and Search Interface

The public-facing home page provides a medicine-and-health search entry point while keeping the research functions available through the navigation structure. It is intentionally separated from the experimental dashboards so that public information retrieval is not confused with model inference.

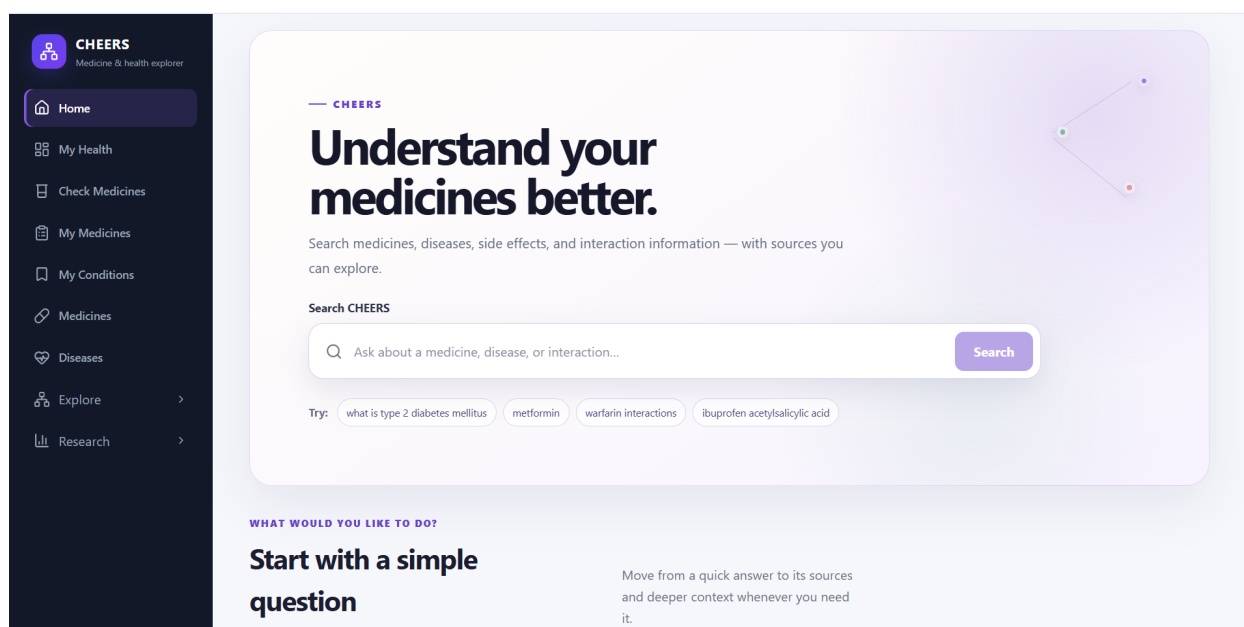


Figure 7: CHEERS public home and search interface.

8.9 DDI Predictor

The predictor accepts an exact selected drug and returns Top-K candidate drugs whose target links are unobserved in the known-positive PrimeKG set. The interface labels the score as a model ranking quantity and avoids terms such as probability or clinical risk.

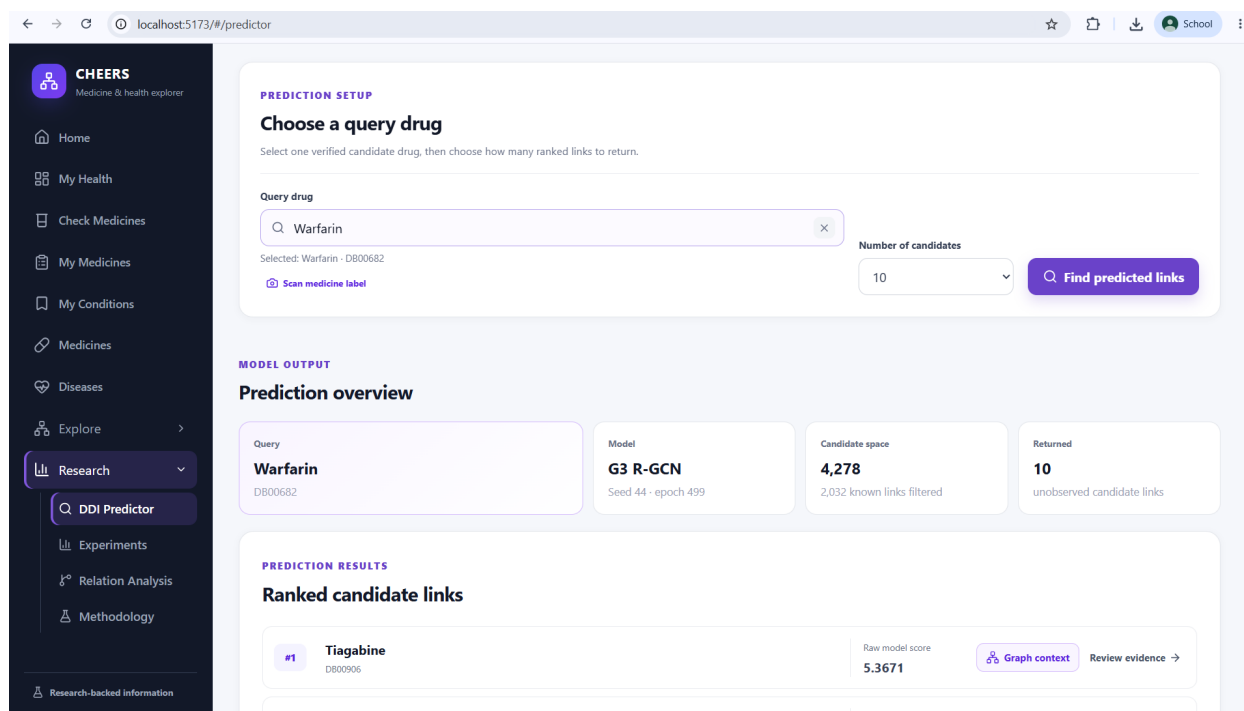


Figure 8: CHEERS DDI Predictor interface showing a selected query drug and ranked candidate-link workflow.

8.10 Pair Graph Explorer

The Pair Explorer loads two drugs by exact project identifiers and retrieves their G3 context. Shared entities are found by intersecting context-node IDs, while preserving the relation labels from each drug to the shared entity.

The view can display:

- shared gene/protein context;
- shared disease context;
- training DDI-neighborhood context;
- entity metadata and relation descriptions.

The explanation text is deterministic and project-scoped, not generated by an LLM. An offline LLM-based explanation layer was considered during development but was not adopted because generated biomedical text could introduce unsupported or hallucinated claims that were not grounded in the stored graph or cited evidence. The final interface therefore uses deterministic relation descriptions and source-grounded metadata and explicitly avoids claiming interaction severity or causal mechanism.

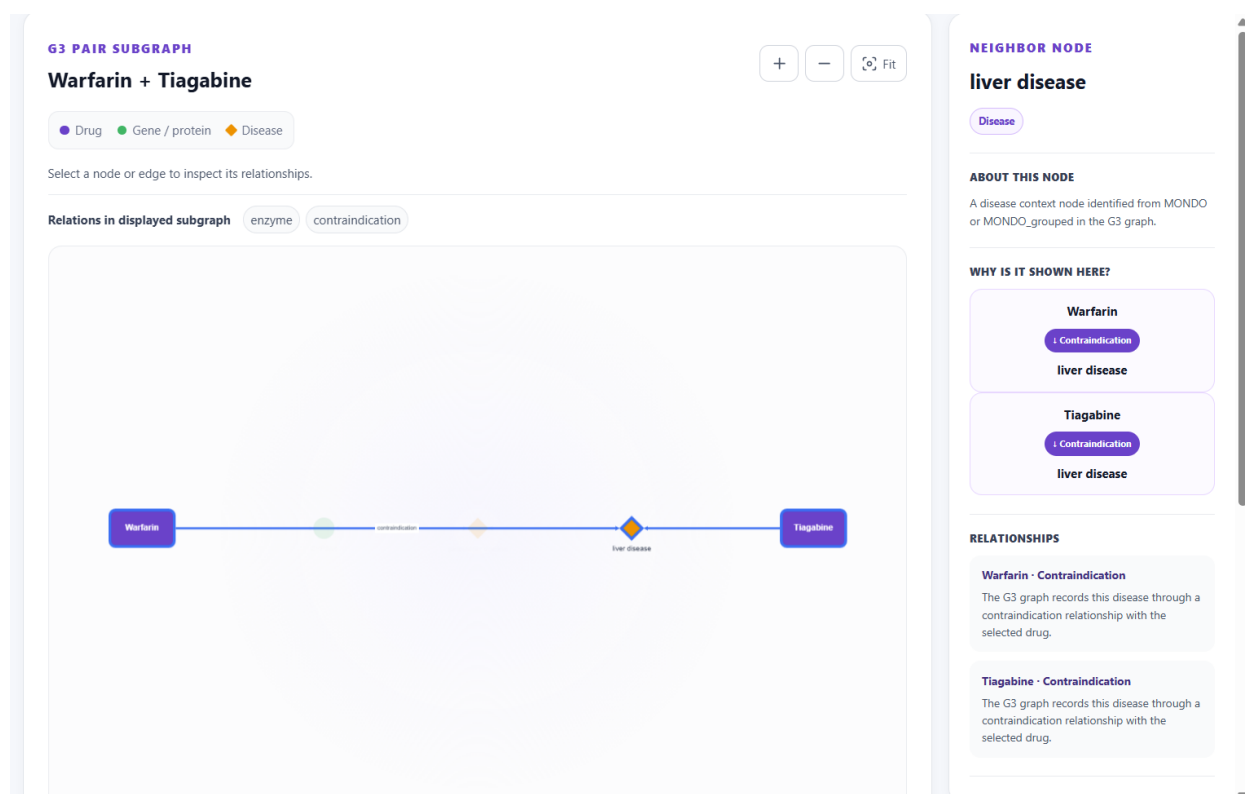


Figure 9: Pair-specific G3 context visualization for Warfarin and Tiagabine, with relation-aware node details.

8.11 Single-Drug Subgraph Explorer

The Subgraph Explorer shows a one-hop neighborhood around one selected drug. Supported relation filters include DDI, Target, Enzyme, Transporter, Carrier, Indication, Contraindication, and Off-label use. Entity filters include Drugs, Gene/Protein, and Diseases.

The default page size is 50 neighboring nodes. Pagination uses offset-based Previous/Next replacement rather than cumulative “load more” appending. This avoids progressively shrinking or overcrowding high-degree visualizations.

Filter changes reset the offset to zero, selected details are cleared on page changes, and the graph is re-fit after each page load.

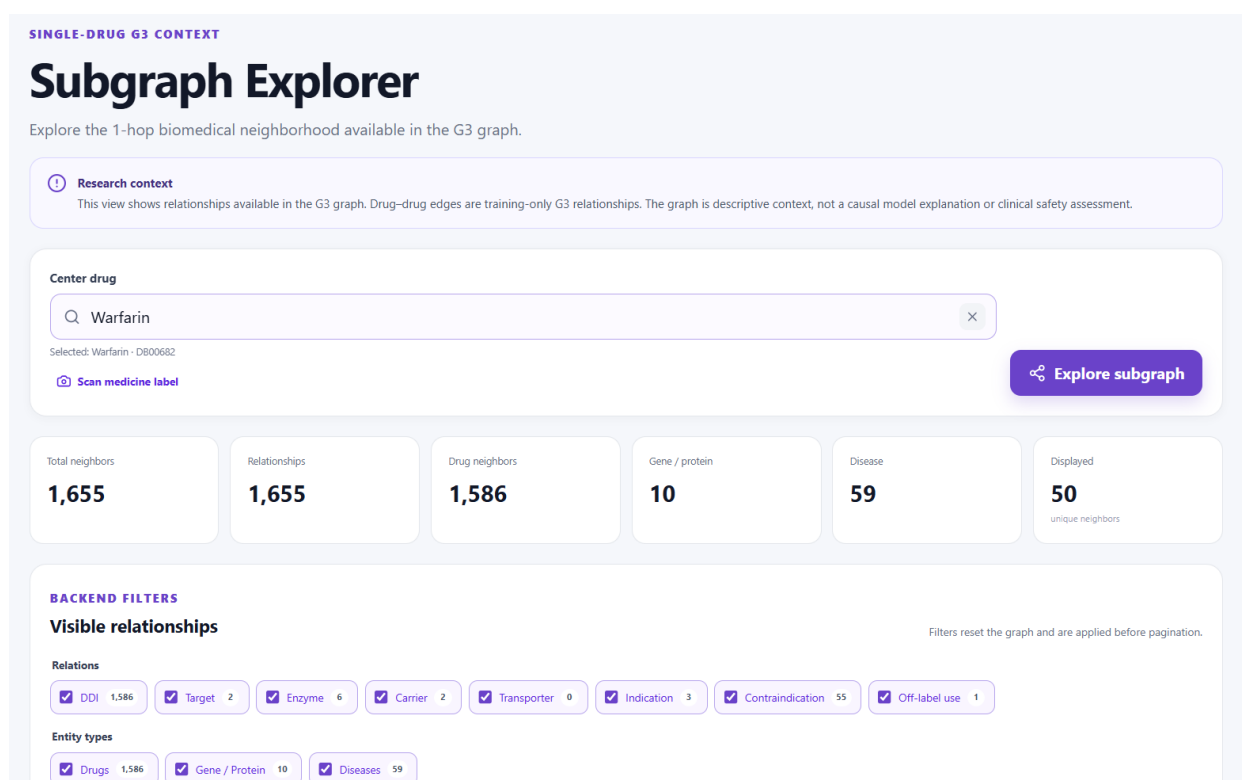


Figure 10: Single-drug one-hop G3 Subgraph Explorer with relation and entity-type filters.

8.12 Entity Details and Grounded Metadata

Selecting a graph node opens a details panel. For gene/protein nodes, matched NCBI Gene metadata can include:

- official symbol;
- official full name;
- organism and taxonomy;
- aliases;
- NCBI summary;
- source and identifier information.

The verified NCBI metadata artifact contains 3,094 human records and uses exact numeric GeneID matching. It does not use fuzzy name or synonym matching for entity attachment.

Drug metadata were grounded using a local ChEMBL 37 database [12]. Where corresponding DrugBank identifiers were available in the project mappings, UniChem [13] was used as an exact identifier bridge to ChEMBL rather than relying on fuzzy drug-name matching. The full local ChEMBL database was used during preprocessing, while only the fields required by the web application were extracted into a compact runtime artifact so that the complete database did not need to be distributed with the deployed system.

Disease metadata were handled separately using verified disease mappings such as MONDO where available. These drug, gene, and disease metadata are application-level grounded information: they were not textual inputs to the R-GCN and do not explain or alter its ranking score.

8.13 Independent FDA and PubMed Evidence

The evidence layer is deliberately separated from model inference.

The FDA/openFDA component uses conservative drug-name/synonym queries to retrieve relevant label sections. PubMed uses NCBI E-utilities with pair-oriented search queries and returns bibliographic metadata such as PMID, title, authors, journal, and date.

The application presents explicit unavailable, no-result, and error states. A missing result must not be interpreted as evidence of safety.

The screenshot displays a web interface titled "What the checked sources returned" with a link to "Open full source view". It is divided into two main sections: "OFFICIAL SOURCE" and "PUBLISHED LITERATURE".

OFFICIAL SOURCE: Official drug-label information

- Card 1:** Titled "Tiagabine" with a "Drug interactions" tag. The text discusses theophylline pharmacokinetics and warfarin interactions. A "Show less" link is at the bottom.
- Card 2:** Also titled "Tiagabine" with a "Drug interactions" tag. The text discusses theophylline pharmacokinetics and warfarin interactions. A "Show less" link is at the bottom.
- Card 3:** Also titled "Tiagabine" with a "Drug interactions" tag. The text discusses theophylline pharmacokinetics and warfarin interactions. A "View label details" link is at the bottom.

At the bottom of the official source section, it states "18 source-backed label mentions retrieved." and provides an "openFDA source" link.

PUBLISHED LITERATURE: Related research articles

- Card 1:** Titled "Gateways to Clinical Trials. June 2002." with the subtitle "Methods and findings in experimental and clinical pharmacology · 2002 Jun". It includes the PMID 12168506 and a link.
- Card 2:** Titled "The importance of drug interactions in epilepsy therapy." with the subtitle "Epilepsia · 2002 Apr". It includes the PMID 11952767 and a link.

Below the literature cards, it states "2 related records retrieved."

Figure 11: Independent evidence view for a drug pair. Official label excerpts and PubMed records are retrieved separately from R-GCN inference and do not alter or validate the model score.

8.14 Medicine Label OCR

The Medicine Label Scanner uses Tesseract.js in the browser with locally bundled worker/core/language assets. The user uploads or captures an image, OCR extracts text locally, and the user can review or edit the extracted text before it is submitted for conservative drug-name matching.

This design has a useful privacy property: the image itself does not need to be uploaded to the backend for OCR. Nevertheless, OCR can misread labels and the matching process is not a pharmaceutical identification system. It is only an input-selection aid.

Only photograph the medicine/package label. Avoid including your name, prescription number, address, or other personal information. The camera is used only to capture the medicine label. Captured images are processed in your browser and are not uploaded or stored by CHEERS.

Use camera Replace photo Remove photo



warfarin.jpg

Possible supported drugs found

► Detected text

Possible supported drugs found

Select one candidate to fill Query drug. Selection does not run a prediction or load graph context.

Warfarin DB00682	Exact label match
----------------------------	-------------------

CHEERS is a research prototype. This scanner only helps select a supported drug and does not provide clinical identification, diagnosis, medication advice, safety assessment, or interaction severity.

Close

Figure 12: Browser-side medicine-label OCR workflow after recognizing printed label text and conservatively matching Warfarin as a supported project drug. OCR is an input-selection aid, not clinical medicine identification.

8.15 Disease and Nutrition Information

The disease-information runtime contains a small set of reviewed educational records. It is an auxiliary web feature and has no effect on R-GCN training or prediction. This separation is important because application breadth should not be mistaken for additional model inputs.

8.16 Research Methodology and Relation-Analysis Views

The application also exposes the controlled G0–G3 design and the verified five-seed relation-ablation results. These pages are presentation layers over frozen research artifacts; they do not rerun training in the browser or backend. The separate DDI-edge cold-start experiment is documented in Chapter 7 and does not have a dedicated web results page.

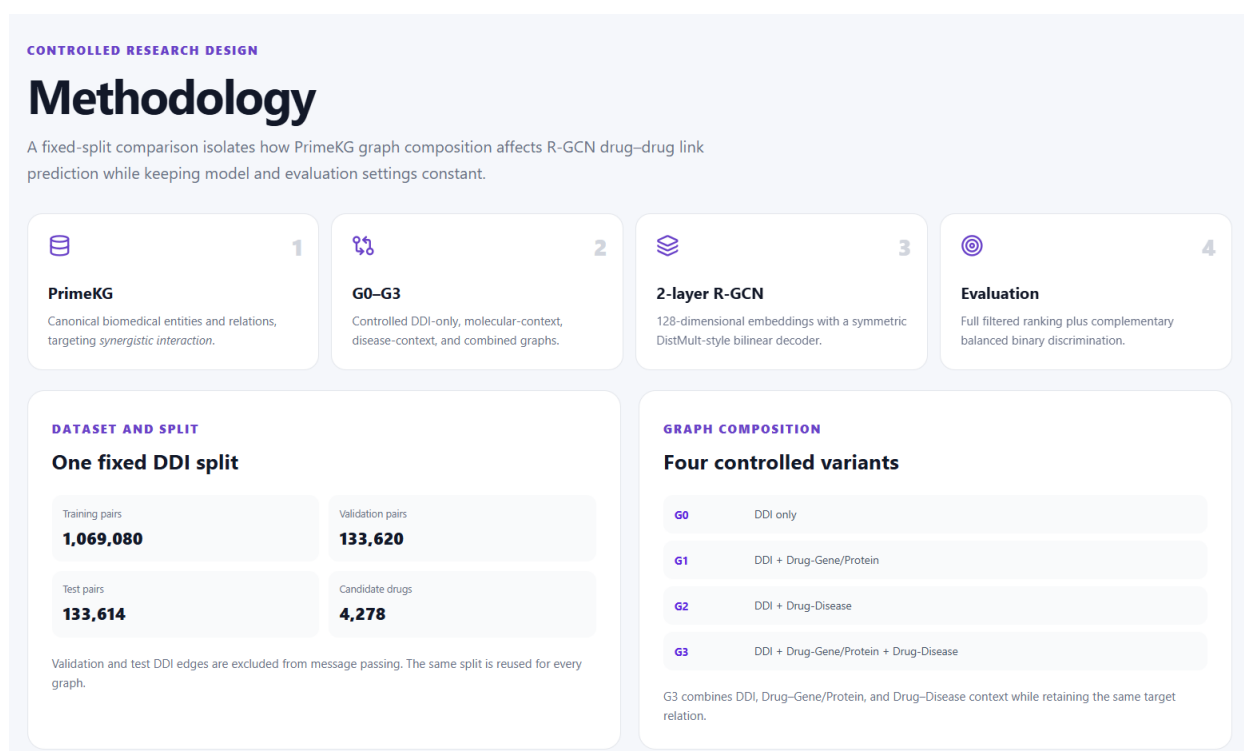


Figure 13: CHEERS methodology page summarizing the fixed split, graph variants, R-GCN architecture, and evaluation design.

ALL OUTCOMES RETAINED

Five-seed relation results

Rank	Relation	Family	MRR mean $\pm$ SD	Mean Δ MRR $\pm$ SD	95% paired interval	Wins vs G0 $\textcircled{1}$	Edges
#1	Target	Drug-Gene/Protein	0.534051 $\pm$ 0.007775	+0.006766 $\pm$ 0.009710	[-0.005290, +0.018823]	$\textcircled{4/5}$	16,380
#2	Carrier	Drug-Gene/Protein	0.529763 $\pm$ 0.008544	+0.002479 $\pm$ 0.005832	[-0.004763, +0.009721]	$\textcircled{3/5}$	864
#3	Indication	Drug-Disease	0.529391 $\pm$ 0.010644	+0.002107 $\pm$ 0.009511	[-0.009703, +0.013916]	$\textcircled{2/5}$	9,388
#4	Transporter	Drug-Gene/Protein	0.528694 $\pm$ 0.009603	+0.001410 $\pm$ 0.005595	[-0.005538, +0.008357]	$\textcircled{4/5}$	3,092
#5	Enzyme	Drug-Gene/Protein	0.523753 $\pm$ 0.006030	-0.003531 $\pm$ 0.006607	[-0.011735, +0.004672]	$\textcircled{1/5}$	5,317
#6	Contraindication	Drug-Disease	0.523467 $\pm$ 0.003972	-0.003817 $\pm$ 0.005689	[-0.010881, +0.003247]	$\textcircled{1/5}$	30,675
#7	Off-label use	Drug-Disease	0.514659 $\pm$ 0.027999	-0.012625 $\pm$ 0.023321	[-0.041582, +0.016332]	$\textcircled{1/5}$	2,568

Figure 14: Five-seed relation-level ablation results shown in the research interface. All displayed 95% paired intervals include zero.

8.17 Responsible Interface Design

CHEERS repeatedly distinguishes four concepts:

1. model ranking score;
2. G3 graph context;
3. grounded entity information;

4. independent FDA/PubMed evidence.

Keeping these categories separate is a core safety and scientific-design principle. The web application is intended to make the research interpretable and auditable, not to transform an uncalibrated link-prediction score into a clinical recommendation.

9. Verification and Reproducibility

9.1 Reproducibility Strategy

CHEERS maintains result summaries, model checkpoints, versioned analysis releases, manifests, checksums, verification scripts, and frozen runtime artifacts. The goal is to distinguish three different reproducibility levels:

1. **runtime reproducibility**: reproduce the selected web inference behavior;
2. **artifact-level analytical reproducibility**: regenerate tables/statistics from preserved result inputs;
3. **full training reproducibility**: rebuild preprocessing and retrain every graph/seed from raw PrimeKG.

The portable repository supports the first two levels more completely than the third.

9.2 Versioned Result Artifacts

The authoritative main five-seed summary and paired validation are stored separately from historical results. Similarly, the five-seed relation-ablation release is versioned and preserves its analysis input, full-precision statistics, tables, figures, and reproduction script.

This separation prevents earlier three-seed findings from being silently treated as current final estimates.

9.3 Checkpoints and Runtime Artifacts

The portable release includes selected graph tensors, drug metadata, selected checkpoints, the G3 lightweight NumPy export, packed known-positive data, and graph-context artifacts needed by the application.

Large raw/preprocessing inputs and all historical training outputs are intentionally not required for serving the application.

9.4 Verification Scripts

Dedicated scripts verify:

- lightweight inference export;
- G3 graph context;
- training DDI neighborhoods;

- NCBI Gene metadata;
- drug/disease description artifacts;
- external evidence/provenance contracts.

Verification is kept separate from user-facing functionality so that a frozen release can be checked independently.

9.5 Manifests and Checksums

Portable-release manifests record expected files and integrity metadata. Selected artifacts use SHA-256 checksums. These checksums help detect accidental corruption, line-ending changes, or replacement of frozen runtime data.

9.6 Git-Based Development History

The repository history documents the progression from main R-GCN experiments to five-seed synchronization, relation ablation, classification, cold-start analysis, DDInter external evaluation, graph explorers, metadata integration, OCR, statistical validation, public-product freeze, and pair-graph UI improvements.

Git history is useful for reconstructing technical evolution, but commit authorship alone is not sufficient evidence for individual thesis contributions. Team-member roles are therefore recorded separately and remain editable until final confirmation.

9.7 Portable Repository Versus University Workspace

The portable repository does not contain every file required to reconstruct the project from raw PrimeKG. Missing or external workspace materials include, among other items:

- the original raw PrimeKG CSV snapshot used by training;
- canonical DDI parquet and split parquet files;
- some full node/relation mappings;
- G0–G2 preprocessing tensors;
- complete original preprocessing and training workspace outputs.

Accordingly, the final thesis should not claim that a fresh clone alone can reproduce every model from raw data. It can reproduce the web runtime and multiple analytical summaries from frozen artifacts, while full retraining requires archived training-workspace resources.

10. Discussion

10.1 RQ1: Does Graph Composition Affect DDI Ranking?

The five-seed results provide evidence that graph composition affects descriptive performance. G3 has the highest mean MRR and the highest mean Hits@1, Hits@5, and Hits@10. G1 also exceeds G0 descriptively in mean MRR, while G2 is slightly lower than G0.

The result therefore rejects the simplistic assumption that all heterogeneous context is equally useful. Molecular-context relations and the fully combined graph behave differently from disease-context-only augmentation.

At the same time, the paired statistical validation limits the strength of the conclusion. G3 wins against G0 in all five matched seeds for the reported ranking metrics, but the 95% paired intervals include zero. The correct interpretation is a small, consistent descriptive advantage under one split, not a statistically established universal improvement.

10.2 RQ2: Which Relation Is Most Useful?

The single-relation ablation does not identify a relation with an established positive effect. Target has the strongest mean improvement and wins four of five seeds, but its paired interval spans zero. Carrier and Transporter show smaller positive means; Enzyme, Contraindication, and Off-label use have negative means.

An important secondary finding is that relation size does not determine usefulness. Carrier is very small but descriptively positive, while the much larger Contraindication relation is negative on average. This suggests that edge semantics, topology, overlap with the DDI structure, and optimization dynamics may matter more than raw edge count.

10.3 RQ3: Does Heterogeneous Context Help Cold Start?

The answer is conditional rather than uniformly positive. In the cold-to-warm condition, G0 is much stronger than G3. In the cold-to-cold condition, G3 retains more signal than G0, although absolute performance is still low.

This result demonstrates why the precise definition of cold start matters. The current setup does not model an unseen drug, and it does not establish a general cold-start solution. It shows only how the two graph variants behave after removing selected DDI edges while keeping nodes known.

10.4 Combined Context Versus Individual Relations

One of the most interesting patterns is that G3 is the highest-mean main graph while no individual add-one relation establishes an improvement. Several explanations are possible:

- multiple relation families may provide complementary context;
- relation interactions may alter neighborhood structure non-additively;
- different relations may help different subsets of drugs;
- seed-level optimization variability may dominate small individual effects;

- the observed G3 mean advantage may itself be a small-sample fluctuation.

The current experiments cannot distinguish these mechanisms, so the present findings are limited to the controlled analyses reported here.

10.5 Ranking and Classification Measure Different Behavior

G3 leads both the main ranking table and the five-seed complementary classification table, but relation-level historical analyses show that ranking and classification can disagree strongly.

This is expected because full ranking asks a relative ordering question over thousands of candidates, whereas classification depends on a threshold and a particular sampled-unobserved distribution. For link prediction, a model may improve the position of true targets without producing a cleaner global binary separation.

10.6 Statistical Uncertainty and Experimental Scale

Five seeds are an improvement over one- or three-seed reporting, but they remain a small sample for inference. More importantly, all seeds use one fixed graph split. The current paired intervals quantify variation due to training randomness on that split only.

The current protocol varies training seeds while retaining one fixed data split; its uncertainty analysis therefore reflects seed-level variability rather than split-level variability.

10.7 External Generalization

The DDInter pilot does not reproduce G3’s internal descriptive advantage. This is valuable because it prevents overgeneralization from the PrimeKG test set. External datasets may differ in drug identity, curation rules, interaction semantics, coverage, and positive-selection mechanisms.

A high-quality external study would require a preregistered mapping protocol, multiple seeds, potentially multiple external resources, and careful treatment of pairs already represented in PrimeKG.

10.8 Interpretability and Evidence Separation

CHEERS deliberately distinguishes graph context from causal explanation. A shared CYP enzyme or disease can help a user understand which graph relationships surround a pair, but it does not prove why the model assigned a score.

The same principle applies to external evidence. FDA labels and PubMed articles can be inspected alongside predictions, but the evidence retrieval process is independent of the R-GCN. This architecture avoids a common interpretability error: presenting post-hoc contextual information as though it were the learned causal basis of a prediction.

10.9 Value of the Web Application

The web application adds practical research value in three ways. First, it exposes frozen model behavior without the original GPU environment. Second, it lets users inspect G3 graph context and relation semantics rather than viewing only a numeric score. Third, it makes the main experiment, relation-ablation results, methodology, and supporting evidence accessible from the same research interface. The cold-start study remains a thesis-level experimental analysis rather than a dedicated web page.

The application is therefore best understood as a research demonstration and audit interface.

11. Project Limitations

The final CHEERS study has several limitations that constrain the interpretation and generalizability of its results: the prediction target is specifically the PrimeKG `drug_drug` relation and therefore does not represent every possible clinical definition of drug–drug interaction; PrimeKG is incomplete and source-dependent, so a pair absent from the known-positive graph cannot be treated as a clinically verified non-interaction; the main G0–G3 experiment is transductive because all evaluated drugs are known graph nodes represented partly by learned node-ID embeddings; the primary comparison uses one fixed train/validation/test split and five training seeds, which measures optimization variability but not variability across alternative data partitions and also gives limited statistical power; the biomedical relation families differ substantially in edge count and node degree, so relation imbalance and drug popularity may affect message aggregation and ranking behavior; training and complementary binary classification use sampled unobserved pairs rather than confirmed negative DDIs; the DistMult-style model outputs are uncalibrated ranking scores rather than interaction probabilities, clinical risk estimates, or severity predictions; shared G3 entities and graph paths are descriptive associations and should not be interpreted as causal biological explanations; the cold-start experiment is a DDI-edge cold-start test on already known nodes rather than a fully inductive unseen-drug benchmark, and its frozen cohort was selected using DDI-degree information that includes the test partition; the DDInter experiment is an exploratory single-seed external diagnostic constrained by conservative drug-identity mapping and therefore is not clinical validation; the portable repository preserves the verified runtime, checkpoints, analysis results, metadata, and verification artifacts but does not contain every raw data and preprocessing asset required to reconstruct the entire university training workspace from the beginning; and application features such as openFDA, PubMed, grounded metadata, disease information, and browser-side OCR can be incomplete, network-dependent, or subject to recognition errors and remain independent of the R-GCN score. Consequently, CHEERS should be interpreted as a controlled academic link-prediction and software-engineering prototype rather than a clinical decision-support, prescribing, diagnosis, or medication-safety system.

12. Conclusion

This project investigated a controlled question: how does biomedical knowledge-graph composition affect R-GCN-based drug–drug link prediction?

The final experiment canonicalized 1,336,314 undirected PrimeKG DDI pairs and used one fixed transductive train/validation/test split across four graph variants. G0 contained only DDI training edges, G1 added Drug–Gene/Protein relations, G2 added Drug–Disease relations, and G3 combined both relation families. All variants used the same two-layer 128-dimensional R-GCN architecture, symmetric DistMult-style DDI decoder, training procedure, candidate drug population, and filtered-ranking evaluation.

Across seeds 42–46, G3 achieved the highest descriptive mean MRR, 0.534209 ± 0.006288 , compared with 0.527284 ± 0.006373 for G0. G3 also achieved the highest descriptive means for Hits@1, Hits@5, and Hits@10. Matched seed differences favored G3 across all five seeds for the reported ranking metrics.

However, every paired 95% interval included zero. For MRR, the mean G3–G0 difference was approximately $+0.006924$, with interval $[-0.002624, 0.016473]$ and exact sign-flip $p = 0.0625$. The project therefore does not claim a statistically established improvement. Instead, it concludes that the combined heterogeneous graph exhibits a small, consistent descriptive advantage under the current five-seed fixed-split protocol.

The five-seed relation-level ablation strengthens the need for cautious interpretation. Target has the largest descriptive mean effect, but all seven relation-level paired intervals include zero. No individual biomedical relation can therefore be identified as reliably beneficial from the present evidence.

Secondary analyses further show that generalization depends on experimental definition. In DDI-edge cold start, G0 is stronger for cold-to-warm queries, whereas G3 retains more signal in the cold-to-cold condition. Exploratory DDInter evaluation does not reproduce G3’s internal advantage. These findings emphasize that performance on one transductive benchmark should not be generalized automatically to unseen drugs, external resources, or clinical settings.

The final research model is integrated into a reproducible application using a verified G3 seed-44 NumPy export, FastAPI backend, and React/Vite frontend. Pair and single-drug graph explorers expose stored G3 context; independent FDA and PubMed services expose source-backed evidence; browser-side OCR supports drug selection; and grounded metadata supports entity identification. These auxiliary features do not enter the R-GCN and do not alter model scores.

The principal scientific lesson of CHEERS is therefore not that a larger biomedical graph is always better. It is that graph composition is an experimental variable whose effect must be measured with controlled data, matched evaluation, uncertainty analysis, and careful interpretation. Under the current protocol, heterogeneous context is promising but not conclusively established as superior. This conclusion is restricted to the controlled fixed-split experiments and supporting analyses reported in this project.

13. Team Contributions

The final CHEERS project was developed collaboratively by all three team members. The principal responsibilities of each member are summarized below. All members also participated in project discussions, application testing, result interpretation, and preparation of the final graduation project materials.

Byambasuren Tuvshinjargal

- Data preprocessing, entity normalization, triple validation, and heterogeneous knowledge-graph construction, including `related_to` data and DDI-pair preparation.
- Implementation, training, and filtered link-prediction evaluation of the exploratory TransE model, including metric analysis.
- Development and execution of the final R-GCN G0–G3 graph-composition training pipeline and primary multi-seed experiments.
- Analysis and interpretation of ranking, classification, cold-start, external-evaluation, and qualitative experimental results.
- Backend/API development, model-runtime integration, frontend development, and integration of research features into the CHEERS web application.
- Application debugging, integration testing, UI/UX refinement, runtime and model verification, and deployment/repository support throughout final system development.
- Research-direction and methodology refinement, related-work analysis, experimental documentation, figures, presentation and poster preparation, and final report preparation.

Galbadrakh Buyandelger

- Processing, normalization, validation, and integration of disease–drug `treated_by` triples into the shared knowledge graph.
- Implementation, training, optimization, and filtered link-prediction evaluation of the exploratory ComplEx model.
- Metric analysis using MRR and Hits@K and comparative analysis of the exploratory TransE, ComplEx, and RotatE results.
- Participation in final R-GCN experimentation, dataset verification, experimental evaluation, and result analysis.
- Application testing, experiment verification, and support for integration of research results into the final CHEERS system.
- Experimental and system documentation, result interpretation, presentation-material preparation, and final report preparation.

Bavuujav Delgerbayar

- Construction and validation of `interacts_with` triples, including duplicate and reverse-pair handling.
- Implementation, training, and evaluation of the exploratory RotatE model, including metric analysis and DDI safety-checking preparation.
- Relation-level ablation, multi-seed performance analysis, five-seed verification, and paired statistical analysis.
- DDI-edge cold-start evaluation and generalization analysis.
- Development of the G3 Subgraph Explorer and graph/context visualization features.
- Construction and integration of local biomedical metadata, entity information, and relationship-detail features into the CHEERS web application.
- Web-application integration, debugging, quality assurance, reproducibility verification, result interpretation, and final report preparation support.

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